

Mining Causality: AI-Assisted Search for Instrumental Variables*

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Abstract

The instrumental variables (IVs) method is a leading empirical strategy for causal inference. Finding IVs is a heuristic and creative process, and justifying their validity—especially exclusion restrictions—is largely rhetorical. We propose using large language models (LLMs) to search for new IVs through narratives and counterfactual reasoning, similar to how a human researcher would. The stark difference, however, is that LLMs can dramatically accelerate this process and explore an extremely large search space. We introduce a *discovery pipeline* that searches for potentially novel and valid IVs using two-step and role-playing prompting strategies. We contend that these strategies simulate the endogenous decision-making of economic agents and social actors and ground language models in real-world scenarios, thereby masking the IV discovery task itself. We apply our method to three canonical areas in economics: returns to schooling,

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demand and supply, and peer effects. We then introduce an *evaluation pipeline* in which we conduct expert surveys and train an LLM judge on the survey data. This evaluation reveals that the IV candidates discovered by our method appear both novel and likely valid, whereas a more direct search tends to return instruments already established in the literature.

Keywords: Causal inference, instrumental variables, exclusion restrictions, artificial intelligence, large language models.

JEL Codes: C26, C36, C5.

1 Introduction

Endogeneity is the major obstacle in conducting causal inference in observational settings. Since the credibility revolution ([Angrist and Pischke, 2010](#)), researchers in social science have developed various identification strategies to overcome endogeneity by restoring versions of quasi-experiments. A leading strategy is the instrumental variables (IVs) method. Over decades, researchers with their ingenuity have discovered IVs in various settings and justified their satisfaction of *exclusion restrictions* (e.g., IVs are conditionally exogenous of latent variables). With its various applicability, the IVs method has prevailed across all subfields of social science and beyond (e.g., [Imbens and Angrist, 1994](#); [Blundell and Powell, 2003](#); [Heckman and Vytlačil, 2005](#); [Hernán and Robins, 2006](#)).¹

Exclusion restrictions are fundamentally untestable assumptions.² Often, in justifying them, researchers resort to *rhetorical* arguments specific to each setting. This non-statistical process follows the discovery of potential candidate IVs, which itself requires researchers’ *counterfactual reasoning* and creativity—and sometimes luck. These elements all contribute to the heuristic processes employed by human researchers.

We demonstrate that large language models (LLMs) can facilitate the discovery of new

¹See [Mogstad and Torgovitsky \(2024\)](#) for a more recent survey.

²An exception is a favorable situation where one enjoys over-identifying restrictions. We discuss this point in our context below. Unlike the exclusion restriction, the IV relevance is testable from data ([Montiel Olea and Pflueger, 2013](#); [Stock and Yogo, 2005](#)).

IVs. Considering that narratives are the primary method of supporting IV exclusion, we believe that LLMs, with sophisticated language processing abilities, are well-suited to assist in the search for new valid IVs and justify them rhetorically, just as human researchers have done for decades. The stark difference, however, is that LLMs can accelerate this process at an exponentially faster rate and explore an extremely large search space, to an extent that human researchers cannot match. It is now recognized that LLMs are capable of conducting counterfactual reasoning—or, perhaps more precisely, exploring alternative scenarios—which makes them a promising tool for causal inference (Jin et al., 2023a,b; Kiciman et al., 2023).

There are at least four benefits to pursuing this machine-assisted approach to discovering IVs. First, researchers can conduct a systematic search at a speedy rate, while adapting to the particularities of their settings. Second, interacting with these tools can inspire ideas for possible domains for novel IVs. Third, the systematic search could increase the possibility of obtaining multiple IVs, which would then enable formal (i.e., statistical) testing of their validity via over-identifying restrictions. Fourth, having a list of candidate IVs would increase the chances of finding actual data that contain IVs or guide the construction of such data, including the design of (natural) experiments to generate IVs.

We introduce a *discovery pipeline* that searches for potentially novel and valid IVs. We show how to construct prompts that guides LLMs to search for candidates for valid IVs. The text representation of exclusion restrictions (among others) is the main component of the prompts. We propose a two-step approach in prompting that divides a discovery task into multiple subtasks, and thus separates counterfactual statements of different complexities. At the same time, we propose using role-playing prompts, arguing that they align with the very source of endogeneity, namely, agents’ decisions.³ By doing so, we equip LLMs with the perspective of agents, enabling them to mimic agents’ endogenous decision-making processes and gather contextual information in realistic scenarios. Another benefit of two-step, role-playing prompts is that they help navigate language models through the realm of real-world

³Decisions of economic agents or social actors have been at the root of challenges for causal analyses (Heckman, 1979; Manski, 1993).

scenarios, rather than anchor them within the narrow realm of academic discourses on IVs. Each stage’s prompt focuses only on a portion of the IV assumptions, framed as an agent’s real-world problem, thereby masking the IV-discovery purpose and maximizing the chance of finding novel IVs.⁴

To prove the proposed concept and illustrate the actual performance of an LLM, we conduct discovery exercises using OpenAI’s ChatGPT-4 (GPT4), one of the leading LLMs, to find IVs in six examples in three well-known areas of empirical economics: returns to schooling, demand and supply, and peer effects. In all six examples, GPT4 produced a list of candidate IVs, some of which appear to be new in the literature and provided rationale for their validity. The list also contains IVs that are popularly used in the literature. In the peer effect example, we also demonstrate that the proposed method can be effective in exploring relatively new topics for empirical research, which may in turn increase the possibility of finding novel IVs.

We then introduce an *evaluation pipeline* based on expert surveys and an LLM judge. In the survey, we elicit validity and novelty scores from empirical economists; these reveal that some of the discovered IV candidates in each domain are novel and at least as valid as the most established instruments in the relevant literature. To assess the discovery strategies systematically, we train and calibrate an LLM judge on the survey data and compare role-playing against direct prompting (i.e., providing the definition of valid IVs and revealing the discovery task), as well as multi-step against single-step approaches. Role-playing yields IVs that appear both novel and likely valid, whereas direct prompting tends to return instruments already established in the literature. Multi-step prompting further improves the search, though the gain over single-step is more modest.

From a broader perspective, the proposal is to systematically “search for exogeneity.” In this spirit, the procedure can be extended to other causal inference methods, such as (i) searching for proper control variables in regression and difference-in-differences analyses and

⁴Even if LLMs circumvent this safeguard and exhibit *memorization* from academic texts, we still find value in the procedure as long as the list of discovered IV candidates includes those that are recognized as new by researchers.

(ii) searching for new running variables in regression discontinuity designs. We construct relevant prompts, apply them to well-known examples in the literature, and successfully identify new variables.

A list of candidate IVs or control variables produced as a result of the proposed method is far from absolute. Rather, it serves as a valuable benchmark that inspires empirical researchers about which types of variables to consider and which domains to explore. The dialogue carried out with LLMs in the process can also help researchers solidify arguments or counterarguments for the validity of variables. After all, artificial intelligence (AI)—like any machines—cannot be the ultimate authority (at least not yet). We believe a human researcher assisted by AI can choose research designs and conduct causal inference more effectively.

We contribute to a growing economics literature on using machine learning and AI to assist research processes. This agenda views these methods as not only data-processing and prediction tools but also as tools that can improve conventional research practices themselves (Mullainathan and Spiess, 2017; Athey and Imbens, 2019; Gentzkow et al., 2019; Dell, 2025; Ludwig et al., 2025). Such practices include the creative and heuristic aspects of research, such as hypothesis generation and empirical design. The use of AI tools (e.g., self-supervised learning, deep learning, generative models) in these stages of scientific discovery has become increasingly common in the natural sciences (see the comprehensive survey in *Nature* by Wang et al., 2023), but remains less common in the social sciences. In interesting work, Ludwig and Mullainathan (2024) and Mullainathan and Rambachan (2024) use generative models to systematically produce hypotheses that are comprehensible by humans in otherwise daunting settings. While we do not explicitly seek to generate hypotheses, the variables we discover implicitly maintain a range of hypotheses about their validity. Regarding the use of foundation models such as LLMs in economic research, Du et al. (2024) use fine-tuned LLMs to predict job transitions and understand career trajectories in labor economics, and Manning et al. (2024) propose to use LLMs to automate the entire process of social scientific research, from data generation to testing causal hypotheses. We employ LLMs in

statistical causal inference by incorporating specific structure from econometric assumptions and allowing for human intervention in discovery processes. A recent survey by Korinek (2023) demonstrates how economists can integrate generative AI as a collaborator throughout their research workflows, including the ideation stage.

This paper also relates to the approach of using LLMs in causal discovery (see Wan et al., 2024, for a recent survey and references therein). However, the fundamental difference of our approach to this line of work is that we use LLMs to systematically discover variables with particular causal structure rather than using LLMs to find causal links among a *pre-determined* set of variables.

The paper is mainly organized in two parts: discovery and evaluation. Part I proposes the discovery pipeline. Section 2 states the IV assumptions, and Section 3 proposes the main idea of IV discovery along with the prompting strategies. Section 4 provides the examples of discovered IV candidates in various domains. Section 5 discusses the use of an adversarial LLM to refine the discovery process. Part II provides the evaluation pipeline and evidence that support the discovery method of Part I. Section 6 contains the expert evaluation of the IV candidates discovered in Section 4. Section 7 then builds an LLM judge and evaluates different discovery strategies, which demonstrates that alternative prompting strategies that disclose the IV-search purpose lead to less novel candidates. It also checks whether datasets are available for the novel candidate IVs discovered. This ends Part II. Section 8 extends the approach to other causal inference settings. Section 9 concludes. The Appendix includes alternative prompting strategies for IV search, additional prompts and findings in the IV discovery examples, and details of LLM judge construction.

2 IV Assumptions

We first formally state our discovery goal. Let Y be the outcome of interest, D be the potentially endogenous treatment, $\mathcal{Z}_K \equiv \{Z_1, \dots, Z_K\}$ be the list of IVs Z_k 's with K being the desired number of IVs to discover, and X be the covariates. Let $Y(d, z_k)$ be the counterfactual

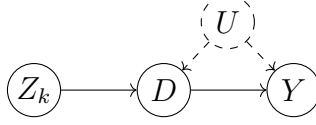


Figure 1: Causal DAG for a Valid IV (X suppressed)

outcome given (d, z_k) . We say Z_k is a valid IV if it satisfies the following assumptions:

Assumption REL (Relevance). *Conditional on X , the distribution of D given $Z_k = z_k$ is a nontrivial function of z_k .*

Assumption EX (Exclusion). *For any (d, z_k) , $Y(d, z_k) = Y(d)$.*

Assumption IND (Independence). *For any d , $Y(d) \perp Z_k$ conditional on X .*

The goal of our exercise is to search for IVs that satisfy Assumptions **REL**, **EX** and **IND**. Suppressing X , Figure 1 depicts the causal directed acyclic graph (DAG) that implies **REL**, **EX** and **IND** and with $Y(d)$ being a transformation of latent confounders U . This diagram is useful in describing our procedures.

3 Prompt Construction

We propose a two-step approach for IV discovery. In Step 1, we prompt an LLM to search for IVs that satisfy a verbal description of **REL** and **EX** (i.e., $(Z_k) \rightarrow (D) \rightarrow (Y)$). In Step 2, we prompt the LLM to refine the search by selecting—among the IVs found in Step 1—those that satisfy a verbal description of **IND** (i.e., $(Z_k) \quad (U)$).⁵ In both steps, the prompts will involve counterfactual statements. In each step, we ask the LLM to provide rationale for its responses. This feature is useful for the user to understand the LLM’s reasoning.

We propose a two-step approach for several reasons: First, LLMs are known to yield better performance when handling subtasks step-by-step, focusing on important details in interpreting the prompts and avoiding errors (Wu et al., 2022). Second, this approach creates

⁵In the Appendix, we present a three-step prompting that focuses on each of **REL**, **EX** and **IND** in each step.

more room for the user to inspect intermediate outputs. In particular, Step 2 involves more complex counterfactual statements than Step 1, allowing the user (and the LLM) to apply varying degrees of attention when fine-tuning is needed. Third, intermediate outputs themselves can provide information and offer insights. Finally, this approach—especially when each step is executed in a separate session—significantly reduces the likelihood that LLMs recognize the task as IV discovery and generate text from relevant academic sources.

Alongside the two-step method, we propose a role-playing approach. When prompts are structured as role-plays, LLMs tend to gather better contextual information, exhibit improved reasoning performances, and generate more tailored and unique responses (Kong et al., 2023). In fact, in most scenarios, the explanatory variable D represents an economic agent’s decision, which naturally facilitate role-playing. Additionally, role-playing prompts are more effective in guiding LLMs to respond as the relevant economic agent rather than as a researcher searching for IVs.

3.1 Step 1: Prompts to Search for IVs

For Step 1, Prompt 1 is a role-playing prompt that queries the search for K_0 IVs (obtaining $\mathcal{Z}_{K_0}$) that satisfy verbal versions of REL and EX (with no X). In all prompts below, each bracketed term represents a user input: [treatment] is the treatment D , [agent] is the economic agent whose decision is D , [scenario] is the specific setting of interest, [outcome] is the outcome Y , and [K_0] is the desired number of variables K_0 . When prompting, we ask the LLM to play the role of [agent] to make a [treatment] decision in a hypothetical [scenario]. Examples of these inputs are given in Section 4.

Prompt 1. *you are [agent] who needs to make a [treatment] decision in [scenario]. what are factors that can determine your decision but do not directly affect your [outcome], except through [treatment] (that is, factors that affect your [outcome] only through [treatment])? list [K_0] factors that are quantifiable.*

explain the answers.

There are at least two variants of Prompt 1 that may be useful in certain scenarios. First, instead of “list [K_0] factors that are quantifiable” one may simply write “list [K_0] factors” or even “list [K_0] factors that are hard to quantify.” This would return candidates of IVs that are harder to measure but can inspire creative data collection (e.g., text, images, or other unstructured data). Second, one can expand Prompt 1 to be more specific about categorizing factors for relevant parties in a given setting. For example, in the schooling scenario (Section 4.1), we request separate lists for student and school factors. This approach can facilitate the evaluation of the results.

3.2 Step 2: Prompts to Refine the Search for IVs

Take the set of IVs, $\mathcal{Z}_{K_0} \equiv \{Z_1, \dots, Z_{K_0}\}$, obtained by running Prompt 1 in Step 1. Next, for Step 2, Prompt 2_x is a role-playing prompt that queries the search for K IVs (obtaining $\mathcal{Z}_K$, $K \leq K_0$) within $\mathcal{Z}_{K_0}$ that satisfy a verbal version of IND. Below, [confounders] is the user input for unobserved confounders of concern and [K] is the user choice of K . In addition, we introduces the notion of conditioning variables in the prompt to reflect covariates X in IND;⁶ role-playing prompts are suitable for this purpose. Although REL also involves conditioning on X , we find that results are not sensitive to a relevant modification of Prompt 1. Prompt 2_x qualifies *both* [agent] and [scenario] by [covariates], the pre-determined user choice of covariates.

Prompt 2_x. *suppose you are [agent] in [scenario] with [covariates]. among the [K_0] factors listed above, choose [K] factors that are most likely to be unassociated with [confounders], which determine your [outcome]. the chosen factors can still influence your [treatment]. for each chosen factor, explain your reasoning.*

⁶The Appendix presents an alternative approach to introducing covariates, namely, jointly searching for IVs and covariates that satisfy REL-IND.

Unlike Prompt 1, this prompt contains a statement about variables typically unobserved to researchers, which may pose challenges. We believe that incorporating the researcher’s prior knowledge on latent confounders helps simplify the overall search process and yield more desirable results. For instance, in the schooling scenario, one can specify “innate ability and personality and school quality.”⁷ In Prompt 2_x, we use the term “unassociated.” If the LLM ever captures the nuance of this word, it reflects the mean independence version of IND, making the search easier.

The recommended approach for incorporating [covariates] is to assign specific values for the covariates. For instance, in the schooling scenario, one can write “suppose you are an asian female high school student from california who considers attending a private college.”⁸ Alternatively, one can simply use terms like “specific” or “particular” along with the name of chosen covariates (e.g., “suppose you are a high school student with specific gender, race, and regional origin who considers attending a college of specific type”).

There are useful variants of Prompt 2_x. First, one can omit [K] and instruct the LLM to “choose all factors” from $\mathcal{Z}_{K_0}$ that are likely to satisfy IND, allowing the LLM to determine K independently; we apply this strategy in all examples later. Second, as a sanity check, one can direct the LLM to select elements in $\mathcal{Z}_{K_0}$ that *violate* IND in addition to those that satisfy it. This can be achieved by adding “also choose factors that are, in contrast, associated with [confounders].” To gain further insights, the user can request explanations for factors that she identifies as valid IVs in the initial set $\mathcal{Z}_{K_0}$ from Step 1, but which are somehow not included in the final set $\mathcal{Z}_K$ by the LLM. We apply the last approach to the application in Section 4.2.1.

⁷An alternative option is to systematically search for possible unobserved confounders.

⁸One can run multiple queries across different covariate values for robustness, although this does not appear to be necessary in most cases unless extreme values are assigned in the initial run.

4 Discovered IV Candidates

Using Prompts 1 and 2_x, we aim to identify candidates for IVs in six domains of three canonical areas in economics: returns to schooling, demand and supply, and peer effects. These examples are chosen for their significance in the literature and commonly employ the IVs method as an empirical strategy. This exercise is to evaluate LLM performance and demonstrate the proposed method’s practicality.

In all the examples, LLMs appear to discover new candidates for IVs and candidates that are related to well-known IVs in the literature. Also, many candidates demonstrate high levels of specificity to their context. We report the full results for the first three examples. For the last three, we only provide summaries and include the comprehensive results in the Appendix. For each example, we present results from the *initial single* run of the prompts without any curation or further refinement. Results across sessions are largely consistent, although they can vary when different values of K_0 and K are chosen. In all the examples, we choose $K_0 = 40$ and let K be determined in the discovery process. We use GPT4 as our LLM.⁹

4.1 Returns to Education

Estimating the causal effects of educational attainment (e.g., college attendance, years of schooling) on earnings has been a significant objective in economics. The main latent confounders in this setting is unobserved individual and school characteristics (e.g., student ability and personality, school quality) that affect both the schooling decisions and future earnings. To address this endogeneity and recover meaningful causal effects (e.g., local average treatment effects (Imbens and Angrist, 1994)), IVs such as distance to schools, tuition fees, and compulsory schooling laws have been widely used in the literature (Card, 1999).

⁹We focus on GPT4 as the other versions of ChatGPT (e.g., GPT4o) produce similar findings. The comparison of prompting strategies in Section 7 uses GPT4o.

4.1.1 Returns to College Attendance

As the first example, we focus on the returns to college attendance. The following is the prompts we use. We explicitly request separate lists for individual and school factors.

Prompt 1.1. *you are a high school graduate. you need to make a college attendance decision. what would be factors (factors of schools and factors of yourself) that can determine your decision but that do not directly affect your future earnings, except through college attendance (that is, that affect your earnings only through college attendance)? list forty factors that are quantifiable, twenty for school factors and twenty for factors of yourself. explain the answers.*

Prompt 2_x.1. *suppose you are a student with family income \$10K per year, who is asian female from california, whose parents have college education, who is catholic. among the forty factors listed above, choose all factors that are not associated with your innate ability and personality and school quality, which determine earnings. create separate lists for school factors and factors of yourself. for each factor chosen, explain your reasoning.*

From running Prompts 1.1 and 2_x.1, the LLM produces candidates for valid IVs and its rationale for the suggestions (Table 1). The list includes IVs that are already established in the literature, such as *distance to college* (Card, 1995, 2001) and *family background* (Card, 1995; Ashenfelter and Zimmerman, 1997), as well as IVs that seem to be new (to our best knowledge): student-related factors as *personal savings for education* and various school characteristics such as *campus crime rates*, *diversity of student body*, *availability of online courses*, *campus housing capacity* and *proximity to urban centers*. The newly found variables

have potential to be valid, especially after being conditioned on covariates. Producing all these results took less than one minute in total.

#	IVs Suggested	Rationale Provided
1	Distance from Home to College (Card (1995), 1955; Card (2001), 3134)	This is purely geographical and independent of personal attributes.
2	Personal Savings for Education	While this might be influenced by a student’s ability to save, it’s largely determined by external economic conditions and family support.
3	Number of Siblings Attending College (Griliches (1977), 2393)	This depends on family decisions and the educational pursuits of siblings, unrelated to the student’s personal capabilities or characteristics.
4	Parents’ Educational Background (Card (1995), 1955; Ashenfelter and Zimmerman (1997), 335)	A demographic factor that precedes the student’s own choices and abilities.
5	Religious Affiliations	A personal and family cultural factor, not necessarily a direct influence on personal academic or career potential.
6	Campus Crime Rates	Safety measures and local crime statistics are external to the student’s attributes.
7	Diversity of Student Body	Reflects the school’s admissions policies and demographic outreach, independent of any one student’s traits.
8	Library Size (Volumes Held)	A resource factor that’s determined by the institution’s funding and priorities.
9	Availability of Online Courses	Indicates the institution’s commitment to flexible learning options, independent of student characteristics.
10	Campus Facilities Quality (Gyms, Labs, etc.)	Dependent on the school’s investment in infrastructure.
11	Campus Housing Capacity	Dictated by the university’s accommodations and facilities, not by student traits.
12	Proximity to Urban Centers	Geographic location of the school relative to cities.
13	Environmental Sustainability Rating	Reflects the institution’s environmental policies and infrastructure.
14	Technology Integration in Classrooms	Indicates the level of modern technological adoption by the school.

Table 1: Returns to College: Suggested IVs and Rationale for Validity

Notes: IV candidates for college attendance are presented. All IV candidates are discovered and explained by GPT4 from a single run of Prompts 1.1 and 2_x.1 with $K_0 = 40$ and K left unspecified. The first 5 rows are categorized by GPT4 to be student-related factors, and the next 9 rows to be school-related factors. When the responses match the IVs described in the literature exactly, we include the corresponding references and their citation counts in parenthesis. The rationale given by LLM can be elaborated further by requesting it in the same session. The total running time was less than 1 minute.

4.1.2 Returns to Years of Education

As a second example, we consider the returns to years of schooling. We omit the prompts as they are similar as before, except that we impose the following role: “you are a student beginning high school in the united states. you need to make a decision on how many more years you will stay in school.”

Table 2 presents the list of candidate IVs (together with the LLM’s reasoning). Among them are instruments well documented in the literature, such as *state education laws* (Angrist and Krueger, 1991; Staiger and Stock, 1997) and *distance to college* (Card, 1995, 2001), alongside several that appear novel and interesting: regulatory and contextual factors such as *credits required for graduation*, *transportation availability*, *local education funding per student*, *local college enrollment rates*, and *local unemployment rates*.

4.2 Demand and Supply

4.2.1 Demand Estimation

Consider estimating demand for consumers in a given market. In estimating the effect of price on demand, the main concern is that price is endogenous as an equilibrium outcome. Researchers have used supply-side IVs that are excluded from the demand equation in a simultaneous system for supply and demand (Angrist et al., 2000) or motivated by structural models (Berry et al., 1995).

Below are the prompts we use, focusing on the setting of Angrist et al. (2000).¹⁰

Prompt 1.2. *you are a dealer at a fish market. you need to set the prices of fish. what would be factors that can determine your decision but that do not directly affect the customers’ demand for fish, except through the price you set (that is, that affect the demand only through fish prices). list forty factors that are quantifiable. explain your answer.*

¹⁰Note that in Prompt 2.2, covariate information is omitted.

#	IVs Suggested	Rationale Provided
1	Credits Required for Graduation	This is a regulatory requirement and does not depend on your personal qualities or the inherent quality of your school but can determine the length of time you stay in school.
2	State Education Laws (Angrist and Krueger (1991), 3951; Staiger and Stock (1997), 10656)	Laws mandating education until a certain age or credits affect all students uniformly, irrespective of individual abilities or school quality.
3	Distance to Nearest College/University (Card (1995), 1955; Card (2001), 3134)	The geographical proximity to higher education institutions is an external factor that doesn't depend on your ability but can influence the decision to pursue further education based on accessibility.
4	Transportation Availability	The ease of getting to and from educational institutions is a logistical concern that affects your ability to attend and continue education, independent of personal academic qualities or school resources.
5	Local Education Funding per Student	This reflects the financial investment in education at a community or state level, impacting the resources available to you, which can influence educational continuation decisions, yet is not a reflection of personal ability.
6	Local College Enrollment Rates	Community trends in college enrollment can create a social environment that encourages further education, which is more about local culture and less about individual capacity.
7	Local Unemployment Rates	Higher local unemployment might incentivize staying in school longer as a way to defer entering a tough job market, a decision influenced by economic conditions rather than personal academic prowess.

Table 2: Returns to Years of Schooling: Suggested IVs and Rationale for Validity

Notes: IV candidates for years of schooling are presented. All IV candidates are discovered and explained by GPT4 from a single run of Prompts 1.1 and 2_x.1 (modified for this example) with $K_0 = 40$ and K left unspecified. When the responses match the IVs described in the literature exactly, we include the corresponding references and their citation counts in parenthesis. The rationale given by LLM can be elaborated further by requesting it in the same session. The total running time was less than 1 minute.

Prompt 2.2. *suppose you are a dealer at the fish market who is selling fish and setting its prices on a day of the week. among the factors listed above, choose all factors are not influenced by fish market conditions or customers' characteristics that determine demand for fish. for each factor chosen, explain your reasoning.*

Table 3 contains the candidate IVs from running Prompts 1.2 and 2.2. There are many *supply-side factors* (e.g., *fuel costs*), which are reasonable candidates for IVs. The *weather conditions* variable used as IVs in Angrist et al. (2000) appears in the list. Interestingly, another supply-side factor, *labor costs*, produced by Prompt 1.2 is not included in the final list produced by Prompt 2.2. When asked “explain why you didn’t include ‘‘labor costs’’ in the final list,” the LLM responded that “labor costs are somewhat flexible and can be adjusted in response to changes in market conditions and customer demand, making them more dynamic than some of the other factors listed.”

4.2.2 Production Function Estimation

Consider estimating a production function that captures the causal relationship between inputs and outputs. The key identification challenge is that input decisions can be correlated with unobserved productivity shocks, which directly influence outputs. To address this, IVs such as input prices have been proposed in the literature (Griliches and Mairesse, 1995), which have been subsequently criticized (Olley and Pakes, 1996; Levinsohn and Petrin, 2003; Akerberg et al., 2015).

In the prompts, we explicitly request separate lists for market factors and firm and manager factors. For this example and all examples below, the actual prompts are contained in the Appendix.

The LLM finds sixteen IV candidates (Table A1 in the Appendix). Notably, IVs commonly suggested in the literature (i.e., *input prices*) do not appear among the LLM’s final

#	IVs Suggested	Rationale Provided
1	Fuel Costs	Costs for fuel typically depend on broader economic factors such as oil prices and are not directly affected by market demand for fish or specific customer characteristics.
2	Fishing Equipment Depreciation	The depreciation of fishing gear like nets and boats is largely a function of use and time, rather than market conditions or customer demand.
3	Weather Conditions (Angrist et al. (2000), 454)	Weather affects fishing operations directly but is not influenced by market conditions or the specific characteristics of customers.
4	Regulatory Costs	Costs associated with licensing and compliance with fishing regulations are dictated by regulatory bodies and do not fluctuate based on customer demand or market conditions.
5	Interest Rates	The rates charged on loans for purchasing boats or equipment are determined by financial markets and policy, rather than direct market conditions of the fish market or customer demands.
6	Insurance Costs	Premiums for insuring fishing equipment and operations are typically based on risk assessments and sector-wide data, rather than being directly influenced by day-to-day market conditions or characteristics of customers.
7	Utility Costs at Sales Points	The cost of utilities like electricity and water at sales points tends to be fixed or based on usage rates that are independent of market demand specifics.
8	Economic Conditions	Broader economic factors that affect overall spending and investment patterns influence operational costs but are not dictated by the fish market conditions or customer preferences.
9	Technological Advances	Investments in technology to improve fishing or sales operations are usually planned based on long-term business strategy and efficiency gains rather than immediate market conditions or specific customer demographics.
10	Government Subsidies	Subsidies to the fishing industry are determined by government policies, which are independent of daily market conditions or customer demands in the fish market.
11	Tariffs on Imports	Tariffs imposed on imported fish are a matter of international trade policy and do not change based on daily fluctuations in market demand or customers characteristics.
12	Employee Training Costs	Costs of training employees in handling and selling fish relate to operational efficiency, which are not directly influenced by the day's market conditions or customer demands.

Table 3: Demand Estimation: Suggested Candidates for IVs

Notes: IV candidates for price are presented. All IV candidates are discovered and explained by GPT4 from a single run of Prompts 1.2 and 2.2 with $K_0 = 40$ and K left unspecified. When the responses match the IVs described in the literature exactly, we include the corresponding references and their citation counts in parenthesis. The rationale given by LLM can be elaborated further by requesting it in the same session. The total running time was less than 1 minute.

selections, echoing similar concerns in the literature (Olley and Pakes, 1996; Levinsohn and Petrin, 2003; Akerberg et al., 2015). Instead, the model suggests factors that may influence input prices, such as *interest rates*, *exchange rates*, *tariffs and trade policies*, *transportation costs*, *real estate prices*, and the firm’s *financial health*, several of which could plausibly be exogenous. There are also other instruments that fall into regulatory factors (e.g., *environmental regulations*) and managerial factors (e.g., *risk tolerance*, *strategic objectives*). The model justifies these by arguing that such variables shape firms’ input choices and long-term positioning but do not directly influence short-run productivity shocks. However, some of these long-term decisions may in fact affect long-term outputs, which may or may not be relevant to the short-run outcomes of concern.

4.3 Peer Effects

Suppose we are interested in the causal effects of peers on an individual’s outcomes within a social network. We consider two well-known examples: (i) the effects of peer farmers on the adoption of new farming technologies (Foster and Rosenzweig, 1995; Conley and Udry, 2010); (ii) the effects of peers on teenage smoking (Gaviria and Raphael, 2001). In both examples, the main source of endogeneity is latent factors that determine the formation of network (e.g., latent homophily). To address this, the literature on peer effects sometimes uses friends of friends as IVs (Bramoullé et al., 2009; Angrist, 2014).

It is worth noting that, in these examples, we prompt the LLM to role-play from the peer’s perspective, which is more convenient than prompting from that of the individual whose outcome is of concern.

4.3.1 Effects of Peer Farmers on New Technology Adoption

Interestingly, some IVs that are suggested in the literature (e.g., *friends of friends*) are not chosen by the LLM (Table A2 in the Appendix). It can be because the model either views them as invalid or is incapable of identifying them. On the other hand, the variable *success*

stories documented and shared seems to relate to the IV used in [Conley and Udry \(2010\)](#), which exploits *variation in the presence of experienced farmers*. Additionally, there are other IVs that seem to be new, notably *access to credit* and *engagement in social media or digital platforms*. Finally, the LLM fails to identify the IV used in [Foster and Rosenzweig \(1995\)](#), namely, *endowed land size*. The validity of this variable is justified in their paper by the specific historical backgrounds of the Indian villages studied. This implies that, for some IVs, providing LLMs with institutional details can be necessary; this can be implemented via LLM’s system messages.

4.3.2 Effects of Peer Teenagers on Smoking Behavior

We choose a teenager in urban Indonesia for its relevance, given that the teenage smoking rate in Indonesia has been recently reported as one of the world’s highest ([Fithria et al., 2021](#)). We consider a social media network in the scenario to illustrate the effectiveness of our approach in exploring relatively recent topics in the literature, thereby highlighting the potential to discover novel IVs.

It is important to note that, given that the prompts are written from the perspective of peers, the variables in the table should be understood as factors influencing *peers* of the focal individual. Given that the setup incorporates modern elements such as social media, we identify many potentially new and interesting IVs, particularly from the social media category such as *engagement rate* and *frequency of posts* (Table [A3](#) in the Appendix). Interestingly, *access to smartphones among peers* can be viewed as a “friends of friends” IV.¹¹

5 Adversarial Large Language Models

One way to refine the responses of an LLM is to request another LLM to play the role of an adversary and review the responses produced by the first LLM, namely the defender LLM. In the adversarial stage, we fully disclose the IV discovery task and ask the adversarial LLM to

¹¹There were other “friends of friends” IVs that are produced from Step 1 but did not survive Step 2.

provide counter-arguments, but without using econometric jargon. These counter-arguments are then given to the defender LLM, which is asked to refine the previous answers. Overall, this process produces more sophisticated responses. We find this to be a useful exercise, which mimics the mental process of a human researcher.

The following is an example prompt given to the adversarial LLM. In the prompt, [the list of variables by Defender] is the list provided the defender LLM, generated from running, for example, Prompts 1–2_x.

Prompt AD1. *you are a researcher who wants to find instrumental variables to estimate the effect of [treatment] on [outcome]. below is a list of candidate instrumental variables. for each variable in the list, provide arguments as to why it may not be a valid instrument:*
[the list of variables by Defender]

Then, the counter-arguments generated from Prompt AD1 are presented to the defender LLM using the following prompt, which follows Prompts 1–2_x.

Prompt 3. *below are counter-arguments for each of your previous answers. based on these arguments, revise your selection and provide a list:*
[the arguments by Adversary]

When running Prompt 3, it is important to encourage the defender to create a list, otherwise, it might become pessimistic and reject all the previous selections.

We apply this adversarial process to our examples in Section 4. Overall, the defender provides more sophisticated answers in the end. In the “Returns to College” example (Section 4.1.1), the variables related to *geographical proximity* or *transportation* have been retained. The other variables related to *school facilities* (e.g., *library size*, *technology integration in classrooms*) have been modified to factors that may better satisfy exogeneity (e.g., *the percentage of renewable energy used on campus*, *the number of green spaces on campus*, *campus*

medical facilities). Interestingly, many variables now appear to be weak IVs, which is sensible. In the “Returns to Years of Education” examples (Section 4.1.2), the *state laws* variable survived, but was renamed to the more descriptive *mandatory minimum years of schooling required by state*. Other variables were also further refined. *GED program availability* emerged. In the “Demand Estimation” example (Section 4.2.1), the *weather condition* variable was refined to *global climate patterns (e.g., El Niño)*, because the counter-argument was that “bad weather can also discourage customers from coming to the market,” which is a compelling argument. Many other variables were refined to reflect *global and macroeconomic conditions*. Again, these variables suffer from being weak IVs or having less individual variation.

6 Expert Evaluation: Discovered IV Candidates

Although evaluating the validity of IVs is intrinsically elusive, we attempt to validate our approach through expert evaluation. To that end, we conducted a survey of experts in various fields of empirical economics—where IVs are heavily used—to elicit their evaluation of the IV candidates discovered by LLMs. We received evaluations from nine experts, each of whom evaluated all 70 IV candidates.

For each of the six domains in Section 4, we collected two metrics through the survey: the IV validity score (1 = least valid, 5 = most valid) and the non-novelty score (1 = least likely to exist in the previous literature, 5 = most likely to exist).¹² Lower non-novelty scores would correspond to *higher* IV novelty. The validity scores should be interpreted with care. First, the difficulty of finding valid IVs varies across domains, and thus the scores should be interpreted relative to a benchmark. We define the benchmark as the average validity of the most established IVs (non-novelty score = 5) in each domain. Second, although the average score may be lower, individual IVs found by LLMs may be viewed as novel and valid. Having these points in mind, for each domain, we report (i) the average validity score given

¹²A similar design is used by Si et al. (2025), who ask expert researchers to score the novelty and feasibility of research ideas generated by LLMs.

Domain [†]	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$ (benchmark)	# of Novel IVs $\geq$ Benchmark ^{††}
1	1.87	2.23	2.72	2.17	2.63	3
2	3.00	2.10	2.39	2.97	3.19	2
3	1.92	2.39	2.63	3.76	3.00	4
4	1.92	2.22	2.70	3.19	3.72	1
5	2.36	2.26	2.96	2.39	3.67	1
6	2.10	2.03	2.53	3.00	2.00	10

Table 4: Average Validity Across Novelty ($k = 1$: most novel; $k = 5$: least novel)

[†]Domain ID: 1. Returns to College Attendance; 2. Returns to Years of Education; 3. Demand Estimation; 4. Production Functions; 5. Peer Effects (technology adoption); 6. Peer Effects (teenage smoking). ^{††}Number of IVs with $k = 1$ or 2 whose validity score is at least the benchmark; these IVs are listed in Table 5.

non-novelty score $E[\text{validity}|\text{non-novelty} = k]$ for each $k = 1, \dots, 5$ and (ii) the names of all IVs such that $E[\text{validity}|\text{non-novelty} = k, \text{IV name}] \geq \text{benchmark}$ for $k = 1$ or for $k = 2$, where $\text{benchmark} = E[\text{validity}|\text{non-novelty} = 5]$.¹³ IVs meeting this criterion can be viewed as relatively novel yet at least as valid as existing ones.

Tables 4–5 report the results. Overall, we observe the following: (i) No IV is perfect; even the most established IVs sometimes averaged a validity score below 3 on our scale.¹⁴ (ii) Finding valid IVs is challenging for both LLMs and human researchers (the latter reflected in the established IVs). (iii) The LLM has the capability of identifying IVs that are viewed as both novel and more valid than the most established benchmarks. (iv) The LLM discovered the greatest number of valid IV candidates in the most novel domain (i.e., the example of peer effects on teenage smoking in social-media networks), which was also the most challenging domain with the lowest benchmark score.

¹³For each IV and each k , validity is averaged over the experts who gave the IV the non-novelty score k . IVs with references in Tables 1–3 (i.e., those verified to be established) are excluded from (ii).

¹⁴The literature recognizes that mildly invalid (or only plausibly exogenous) IVs are ubiquitous (Conley et al., 2012; Deaner, 2025).

Domain [†] (Benchmark)	Novel IVs Suggested
1 (2.63) ^{††}	Campus Housing Capacity (+0.70) ^{†††} Technology Integration in Classrooms (+0.37) Campus Crime Rates (+0.03)
2 (3.19)	Credits Required for Graduation (+0.31) Transportation Availability (+0.31)
3 (3.00)	Fuel Costs (+1.00) Tariffs on Imports (+1.00) Technological Advances (+0.00) Government Subsidies (+0.00)
4 (3.72)	Exchange Rates (+0.28)
5 (3.67)	Access to Credit (+0.33)
6 (2.00)	Access to Smartphones Among Peers (+1.50) Content Reach (+1.33) Engagement Rate (+1.00) Local Regulations on Tobacco Advertising (+1.00) Number of Platforms Used (+0.40) Frequency of Posts (+0.33) Visibility at School Events (+0.25) School Policies on Smoking (+0.00) History of Disciplinary Actions at School (+0.00) Academic Performance (+0.00)

Table 5: IVs with $k = 1$ or 2 Outperforming Benchmark ($k = 1$: most novel; $k = 5$: least novel)

[†]Domain ID: 1. Returns to College Attendance; 2. Returns to Years of Education; 3. Demand Estimation; 4. Production Functions; 5. Peer Effects (technology adoption); 6. Peer Effects (teenage smoking). ^{††}The validity score of benchmark is in parenthesis. ^{†††}The improvement of validity score over benchmark is in parenthesis.

7 LLM Judge Evaluation: Comparison to Direct Approaches

7.1 Possible Alternative Approaches

One alternative to the proposed prompting strategy (i.e., two-step, role-playing prompts) would be to explicitly mention the term “instrumental variables” and inform LLMs that the ultimate objective is to “search for instrumental variables.” Additionally, one can verbally train LLMs to understand the IV assumptions before conducting the search. Unfortunately, we find that this direct approach tends to suffer from *memorization*, where LLMs primarily focus on sources from existing academic articles and textbooks relevant to IVs. One can easily verify that, when applying prompts from the direct approach to the examples in Section 4, GPT4 generates answers that are reminiscent of established IVs in the literature. Moreover, when additionally prompted with “did you have sources for your previous responses?,” GPT4 responds by citing well-known references in each application that propose IVs. In contrast, when the same question is asked after the proposed procedure, GPT4 responds that it “didn’t have specific sources.” When pressed to provide sources from the web, in the returns to education application, for example, it cites academic references from behavioral and social sciences on students’ decision-making, including qualitative and case studies.

Motivated by this, we compare the performance of the following five strategies to understand the contribution of each aspect of the proposed approach:

1. Role-playing and two-step (the proposed approach);
2. Role-playing and one-step;
3. Direct and two-step;
4. Direct and one-step (the direct approach discussed above);

5. Direct and one-step with an emphasis on novelty.

The one-step role-playing strategy merges Prompts 1 and 2_x into a single prompt. The direct strategies first provide the LLM with the definition of a valid IV and then ask for IVs in the given setting, either in one step or in two steps that mirror Prompts 1 and 2_x; the last strategy additionally asks for IVs that have not been used in the existing literature. The prompts are given in Appendix E.1. The comparison of strategies 1 vs. 3 and 2 vs. 4 would reveal the effect of the role-playing prompt on performance, while the comparison of 1 vs. 2 and 3 vs. 4 would show the effect of the two-step prompt.

For each of the six domains in Section 4, we run each strategy ten times to account for the stochasticity of LLM responses.¹⁵ Each run generates twenty IV candidates. We then score each discovered IV candidate in terms of validity and non-novelty. There are in total 6000 (= 5 (strategies) × 6 (domains) × 10 (runs) × 20 (IVs)) candidates to score, and so conducting an additional expert survey for them would be exceedingly expensive. Therefore, in what follows, we introduce an evaluation pipeline that draws a more systematic comparison of the alternative strategies at significantly lower cost.

7.2 Construction of LLM Judge

We develop a procedure to build an *LLM judge* trained and calibrated to the small-scale human expert survey we conducted in Section 6.¹⁶ Specifically, the LLM judge is an LLM that plays the role of an econometrician. It is instructed that it is an academic economist evaluating candidate IVs, and it scores each candidate on the same two scales as the experts, validity and non-novelty. Its scores are then calibrated to predict the experts' scores. The judge is only a proxy for expert judgment, but it is substantially cheaper and more scalable than a survey. The construction consists of three steps; the prompts and further details on the construction and evaluation of the LLM judge are given in Appendix E.2.

¹⁵We use GPT4o with temperature 0.7 and fix $K = 20$ in all strategies so that they are compared on an equal footing; the candidates therefore differ from those evaluated in Section 6.

¹⁶In addition to the nine experts in Section 6, we use the scores of two experts whose validity scores were incomplete and who were thus dropped in Section 6.

Step 1. To account for the small sample size of the expert survey, we denoise the expert scores by hierarchical Bayesian shrinkage (Li and Tobias, 2011).¹⁷ Each expert was asked to score every IV in the survey, but the experts differ in leniency and the domains differ in difficulty. For each of the validity and non-novelty scores, we therefore estimate an ordered-logit model in which the score of IV i by expert j depends on the latent quality of the IV, $\theta_i = \alpha_{d(i)} + u_i$ (where $\alpha_{d(i)}$ is a domain effect and u_i is an IV-specific deviation) and on the leniency of the expert, b_j . Specifically, the probability that expert j gives IV i the score $s \in \{1, \dots, 5\}$ is $\Lambda(\kappa_s - \theta_i - b_j) - \Lambda(\kappa_{s-1} - \theta_i - b_j)$, where Λ is the logistic distribution function and $-\infty = \kappa_0 < \kappa_1 < \dots < \kappa_4 < \kappa_5 = \infty$ are cutoffs. The model is estimated by Markov chain Monte Carlo with Gaussian priors on α_d , u_i , and b_j whose variances are estimated from the data.¹⁸ The judge is then calibrated to predict $\hat{m}_i \in [1, 5]$, the posterior mean of the expected score of IV i by a typical expert (i.e., with $b_j = 0$).

Step 2. We prompt the judge to evaluate candidate IVs. The system message instructs the LLM that it is an academic economist evaluating candidate IVs, explicitly states the definition of a valid IV (i.e., verbal versions of REL-IND), and gives the definitions of the two scores, validity and non-novelty, closely following the survey questionnaire.¹⁹ We give the LLM the domain information (i.e., the setting, the treatment decision, and the source of endogeneity) and implement *few-shot learning* (Brown et al., 2020) by providing, as few-shot examples, all but one of the expert-scored IVs of the same domain with their denoised scores (i.e., $N_d - 1$ examples, where N_d is the number of expert-scored IVs in domain d). It is then asked to score the remaining expert-scored IV in that domain.²⁰ We use GPT5 as the judge, with one independent query per candidate.

Step 3. We calibrate the LLM judge by mapping the judge’s raw score $k \in \{1, \dots, 5\}$ to the

¹⁷One can use empirical Bayes instead, although the hyperparameters can be poorly estimated given the small sample size.

¹⁸The prior on u_i shrinks the latent quality of an IV with few or noisy scores toward the mean of its own domain (rather than the overall mean).

¹⁹To be consistent with the expert survey, we ask the judge to score the IV itself rather than the quality of its rationale.

²⁰Restricting the examples to the candidate’s own domain keeps the scores relative to the domain, as in the survey, and fixes the set of examples, so that every candidate of a domain is scored with an identical prompt.

expert scale by a monotone calibration function $\hat{g}(k)$. Among affine, isotonic, and ordered-logit functions, leave-one-out cross-validation selects the affine mapping for validity and the ordered-logit mapping for non-novelty.

We evaluate the LLM judge using the hold-out IVs in the expert survey that are removed from the few-shot examples. Figure 2 plots the judge’s scores against the denoised expert scores. Unlike its raw scores (left panels), the judge’s calibrated scores (right panels) track the experts reasonably well. As one would expect, the judge is better at telling whether an IV is established in the literature than at scoring its validity, which is the more elusive task for human experts as well.

Once calibrated and evaluated, the judge is fixed and applied to all candidate IVs produced with the five strategies, using the full set of expert-scored examples of each domain (i.e., all N_d expert-scored IVs of the domain). Because $\hat{g}$ is strictly increasing, the calibrated scores take five values per dimension: $\hat{g}(1), \dots, \hat{g}(5)$ are 1.95, 2.34, 2.73, 3.11, and 3.50 for validity and 1.29, 1.92, 2.32, 2.99, and 3.64 for non-novelty, so that a calibrated score at or above $\hat{g}(k)$ means that the raw score is at least k . In what follows, we call a candidate *novel* if its non-novelty score is $k = 1$ or 2, and, as in Section 6, we compare validity with a benchmark. Since the judge rarely assigns $k = 5$, we define the benchmark of a strategy in a domain as the average calibrated validity of that strategy’s candidates with $k = 4$ or 5 in the domain.

7.3 Performance Comparison

Using the LLM judge, we produce validity and non-novelty scores for all candidate IVs and compare the strategies. Table 6 reports summary statistics of the scores by strategy. Overall, the direct strategies produce candidates that are on average more valid but much less novel than those of the role-playing strategies. Their average validity is higher by 0.18–0.26 on the 1–5 scale (less than one step of the raw scale), but only 7–18% of their candidates are novel, compared with a third of the candidates of the role-playing strategies; conversely, 41–51%

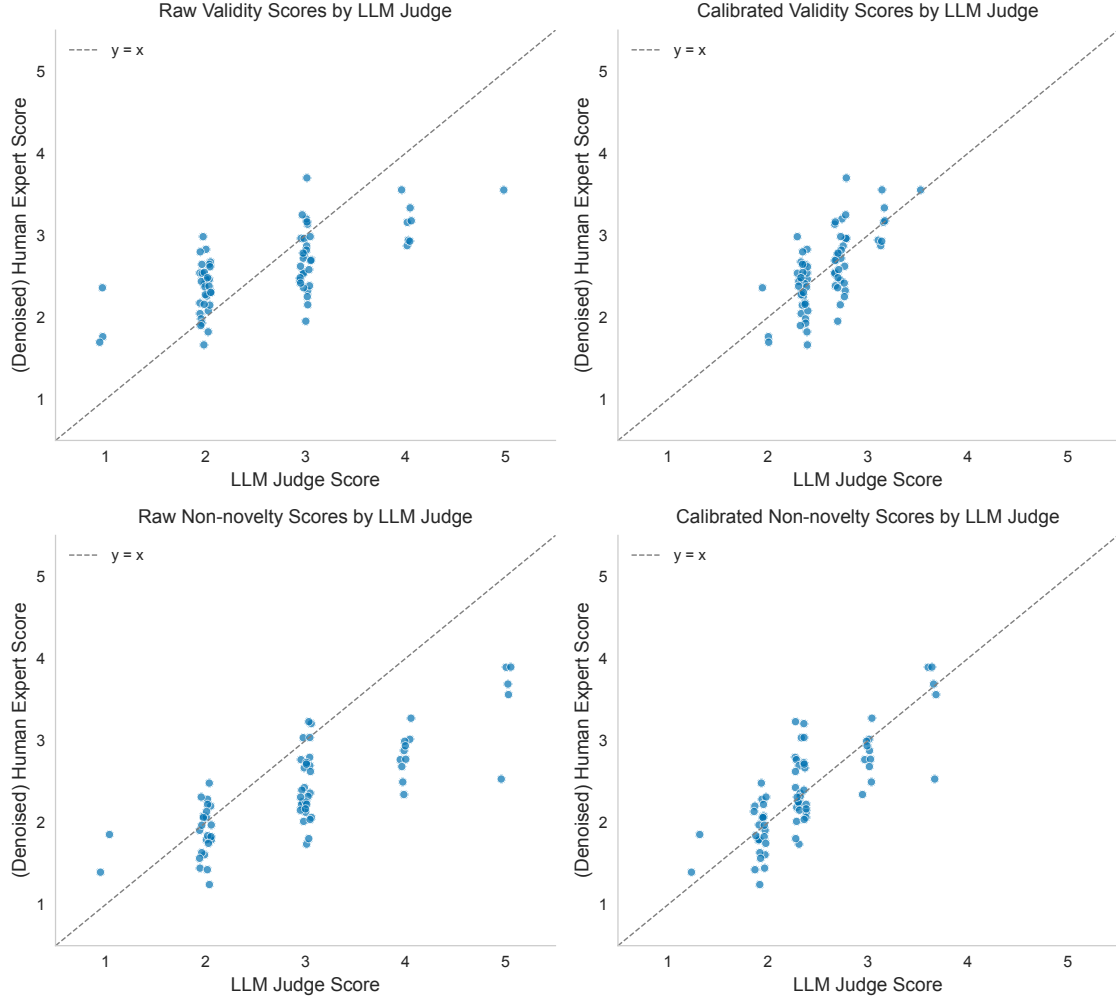


Figure 2: LLM Judge versus Human Expert Scores of the Survey IVs

Notes: Each point is one of the 70 IVs in the expert survey of Section 6. The horizontal axis is the score of the LLM judge, raw (left panels) or calibrated by $\hat{g}$ (right panels); the vertical axis is the denoised expert score $\hat{m}_i$. Top panels: validity; bottom panels: non-novelty. The dashed line is the 45-degree line. Points are jittered horizontally for visibility.

Strategy [†]	<i>N</i>	Validity		Non-novelty		Share of candidates (%) ^{††}	
		Mean	SD	Mean	SD	Novel	Established
Two-step role-play	1200	2.54	0.28	2.35	0.44	34	23
One-step role-play	1200	2.56	0.29	2.35	0.42	31	21
Two-step direct	1200	2.74	0.29	2.59	0.49	18	44
One-step direct	1200	2.77	0.30	2.68	0.45	7	51
One-step direct (novelty-encouraged)	1200	2.80	0.30	2.58	0.42	9	41

Table 6: Summary Statistics of Calibrated Validity and Non-novelty Scores by Strategy

[†]Strategies correspond to the numbered list in Section 7: 1. two-step role-play; 2. one-step role-play; 3. two-step direct; 4. one-step direct; 5. one-step direct with an emphasis on novelty. *N* is the number of scored candidates (6 domains $\times$ 10 runs $\times$ 20 candidates; one-step role-play runs whose answer was parsed into more than 20 rows are capped at their first 20 candidates after removing rows that merely repeat a candidate’s explanation). Scores are the calibrated scores of the LLM judge (Section 7). ^{††}Shares of candidates with non-novelty score $k = 1$ or 2 (novel) and $k = 4$ or 5 (established).

of their candidates are scored as established IVs, compared with 21–23% (Table 6). This is consistent with the memorization discussed in Section 7.1.

Table 7 reports, for each strategy and domain, the average calibrated validity across non-novelty scores in the format of Table 4, together with the number of novel candidates that are at least as valid as the benchmark. These candidates are listed in Tables A4–A7 in Appendix E, the analogs of Table 5. Because a strategy often proposes the same IV under several names across runs (e.g., *campus safety* and *campus safety statistics*), we group the candidate IVs of each domain into *instrument families*, namely clusters of candidates whose meanings are close (measured by text embeddings), and we count and list families rather than names.²¹

As in the expert evaluation, validity tends to increase with non-novelty within each strategy (Table 7). Among novel candidates, however, the average validity is similar across strategies (about 2.5), so the higher overall validity of the direct strategies reflects their larger share of established candidates. There are also exceptions where the novel candidates are on average more valid than the established ones: the domain of peer effects on teenage smoking for both role-playing strategies and, for the two-step role-playing strategy, the domain of returns to years of education.

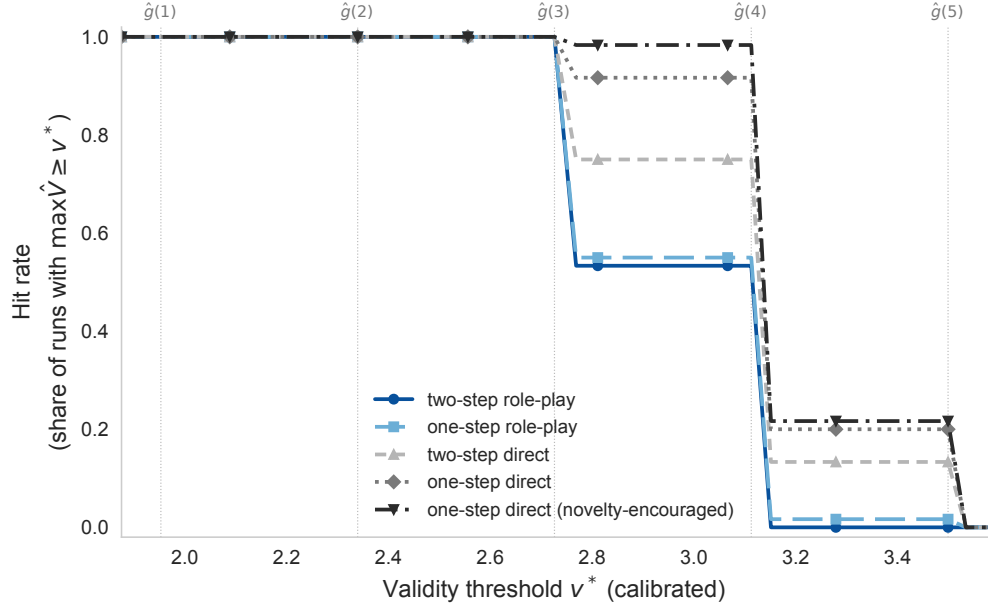
Importantly, the proposed strategy discovers far more IV candidates that are novel yet at least as valid as the benchmark than the other strategies: 60 families, compared with 26

²¹Similarly, Zhang et al. (2025) evaluate the diversity of LLM outputs by partitioning repeated generations into classes of functionally equivalent responses and counting the distinct classes.

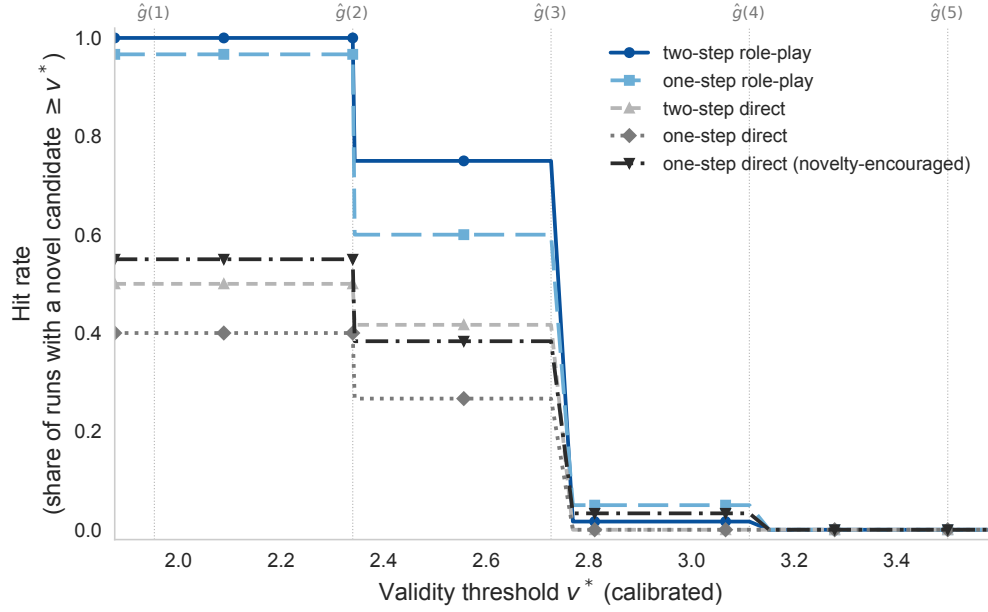
Domain [†]	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$	Benchmark ^{††}	# of Novel IVs $\geq$ Benchmark ^{†††}
<i>Two-step role-play</i>							
1	2.34	2.38	2.33	2.47	2.84	2.59	1
2	–	2.33	2.39	2.30	2.15	2.28	23
3	–	2.68	2.76	2.97	–	2.97	0
4	–	2.42	2.66	2.89	–	2.89	0
5	–	2.49	2.50	2.53	–	2.53	9
6	–	2.55	2.41	2.32	–	2.32	27
<i>One-step role-play</i>							
1	2.34	2.39	2.38	2.56	2.90	2.71	3
2	–	2.30	2.35	2.43	2.47	2.43	0
3	–	2.67	2.74	2.94	–	2.94	0
4	–	2.34	2.65	2.89	–	2.89	0
5	–	2.49	2.56	2.61	–	2.61	8
6	–	2.51	2.41	2.38	–	2.38	15
<i>Two-step direct</i>							
1	–	2.73	2.72	2.88	3.12	2.97	0
2	–	2.46	2.55	2.74	3.12	2.88	0
3	–	2.63	2.84	3.00	–	3.00	0
4	–	–	2.81	2.90	–	2.90	0
5	–	2.53	2.57	2.67	–	2.67	2
6	–	2.52	2.42	–	–	–	–
<i>One-step direct</i>							
1	–	2.73	2.66	2.88	3.14	2.98	0
2	–	2.44	2.54	2.65	3.14	2.79	0
3	–	2.47	2.80	2.99	3.11	2.99	0
4	–	–	2.85	2.93	–	2.93	0
5	–	2.34	2.55	2.65	–	2.65	0
6	–	2.48	2.73	2.98	–	2.98	0
<i>One-step direct (novelty-encouraged)</i>							
1	–	2.58	2.72	2.91	3.16	2.97	0
2	–	2.44	2.53	2.82	3.06	2.88	0
3	–	2.49	2.87	3.13	3.31	3.13	0
4	–	2.73	2.89	2.97	–	2.97	0
5	–	–	2.71	2.78	3.11	2.79	0
6	–	2.46	2.78	2.73	–	2.73	4

Table 7: LLM Judge: Average Calibrated Validity Across Novelty by Strategy ($k = 1$: most novel; $k = 5$: least novel)

[†]Domain ID: 1. Returns to College Attendance; 2. Returns to Years of Education; 3. Demand Estimation; 4. Production Functions; 5. Peer Effects (technology adoption); 6. Peer Effects (teenage smoking). ^{††}Benchmark = $E[\text{validity} \mid \text{non-novelty} \geq 4]$: because the LLM judge rarely assigns $k = 5$, the benchmark pools $k = 4$ and 5 (cf. $k = 5$ in Table 4). Cells are averages of calibrated validity over all candidates in the bin, pooled across the 10 runs; “–” marks an empty bin. Non-novelty bins are the judge’s raw scores. ^{†††}Number of instrument families with $k = 1$ or 2 whose average calibrated validity is at least the benchmark, as listed in Tables A4–A7 in Appendix E; “–” where the benchmark is undefined. Strategies correspond to the numbered list in Section 7: 1. two-step role-play; 2. one-step role-play; 3. two-step direct; 4. one-step direct; 5. one-step direct with an emphasis on novelty.



(a) All candidates



(b) Novel candidates only ($k = 1$ or 2)

Figure 3: Hit-Rate Curves by Strategy

Notes: The hit rate at threshold v^* is the share of runs (60 per strategy: 6 domains $\times$ 10 runs) containing at least one candidate whose calibrated validity is at least v^* ; panel (b) counts only candidates with non-novelty score $k = 1$ or 2 . Vertical lines mark $\hat{g}(k)$, the calibrated value of the judge's raw score k , so the hit rate at $\hat{g}(k)$ equals the share of runs with at least one candidate whose raw validity score is at least k . Strategies correspond to the numbered list in Section 7: 1. two-step role-play; 2. one-step role-play; 3. two-step direct; 4. one-step direct; 5. one-step direct with an emphasis on novelty.

for the one-step role-playing strategy and at most 4 for the direct strategies (last column of Table 7). As in Section 6, these IVs concentrate in the domains with the lowest benchmarks, most of all peer effects on teenage smoking, which is viewed as the most novel domain.²²

Finally, Figure 3 reports, for each strategy, the *hit rate*, namely the share of runs that produce at least one candidate whose calibrated validity is at least a threshold v^* , for all candidates (panel (a)) and for novel candidates only (panel (b)).²³ The hit rate is a natural metric for a discovery tool, because a researcher needs only one valid IV from a run to proceed. The hit rates lead to the same conclusion from the researcher’s perspective. Every strategy produces a candidate with a raw validity score of at least 3 in every run, and at higher thresholds the direct strategies have higher hit rates because they reproduce established IVs (Figure 3(a)). However, once the candidate is required to be novel, the ranking is reversed: the proposed strategy produces a novel candidate with a raw validity score of at least 3 in 75% of the runs, the one-step role-playing strategy in 60%, and the direct strategies in 27–42% (Figure 3(b)).

To understand the separate effects of role-playing and two-step, we compare strategies 1 vs. 3 and 2 vs. 4 for the former and strategies 1 vs. 2 and 3 vs. 4 for the latter. Both role-playing and two-step raise the share of novel candidates (Table 6), the number of novel candidates above the benchmark (Table 7), and the hit rate for novel candidates (Figure 3(b)). However, role-playing dominates the improvement: for example, the hit rate for novel candidates with a raw validity score of at least 3 increases by 33 percentage points with role-playing and by 15 percentage points with two-step. Because role-playing is introduced to mask the IV discovery task, its effect includes the effect of masking. Notably, explicitly asking for novel IVs in the direct prompt has only a modest effect overall: the share of novel candidates rises only slightly (Table 6), the novelty-encouraged strategy finds few novel

²²Note that the benchmark is that of each strategy, and it is low for the role-playing strategies in these domains (about 2.3, i.e., a raw validity score of about 2), so that many of these families are novel candidates of modest validity; a comparison with a common threshold is given by the hit rates below.

²³The hit rate is analogous to the *pass@k* metric that Chen et al. (2021) use to evaluate LLM code generation, namely the share of problems for which at least one of the generated code samples passes the unit tests.

candidates above the benchmark (Table 7), and the hit-rate improvement is not substantial (Figure 3).

We believe that this additional evidence shows the effectiveness of the proposed pipeline for IV discovery. Importantly, we view this evaluation pipeline as part of the overall pipeline that empirical researchers can use in IV discovery tasks. The above exercises provide useful implications for the researchers to refine the search outputs. First, running multiple sessions and collating IV candidates into families can be helpful. This is especially the case given the correlation of LLM responses. Second, given this aspect, one would want to identify candidates that are scored as both novel and valid by the LLM judge across families (as in Table 7) or runs (as in Figure 3).

7.4 Data Availability for Candidate IVs

One may wonder whether this discovery exercise remains only rhetorical without being grounded in real data. We believe that the availability of data is not the most important criterion in the search process, because the discovery exercise can inspire researchers to design experiments and collect novel datasets. Nonetheless, it would be useful to check whether the IV candidates proposed by LLMs have corresponding datasets readily available online.

To that end, we take all candidate IV names from the two-step role-play strategy that were scored by the LLM judge as both novel ($k = 1$ or 2) and at least moderately valid (raw validity score of at least 3) at least once across the ten runs. This gives 87 distinct names across the six domains, which the family clustering groups into 47 instrument families.

For each candidate, we ask an LLM with web-search access (GPT5) to locate an actual dataset that contains a measure of the candidate together with the treatment, outcome, and covariates of the domain, either in a single source or in two sources that can be linked. The LLM reports the web page of the dataset and how to obtain the data, or else reports that no such dataset was found.²⁴ Tables A8–A13 in Appendix E.3 list the results by domain.

²⁴The LLM is instructed never to invent a dataset or a web address, and we check automatically whether

A dataset was found for 30 of the 47 IV families: 4 of 4 families for returns to college, 1 of 1 for returns to years of education, 8 of 12 for demand estimation, 5 of 7 for production functions, 5 of 9 for peer effects on technology adoption, and 7 of 14 for peer effects on teenage smoking.²⁵ Data are scarcest in the most novel domain, peer effects on teenage smoking in social-media networks, where many candidates describe features of online behavior that existing surveys do not measure. Of the 42 datasets reported for individual candidate names, an independent check by a second LLM, which visited the reported web pages, confirmed 32, judged 8 plausible, and refuted 2; in most cases the candidate is measured by a proxy rather than directly. We view these results as encouraging: for a majority of the novel and valid candidates, an empirical analysis can start from existing data. If finding datasets is the ultimate goal, then one may want to increase the number of IV candidates the discovery pipeline produces (which is 20 in the current exercise) to maximize the chance. One can also consider including data availability as an additional criterion for the adversarial LLM discussed in Section 5.

Note that we do not impose the data availability criterion directly in the search prompt for two reasons. First, it is difficult to impose such a criterion when the LLM plays the role of an economic agent rather than a researcher. Second, imposing it would discourage the LLM from finding truly novel, unexplored IV candidates. Nonetheless, integrating the post-search data-availability check into the proposed pipeline of discovery and evaluation can still make the pipeline relevant to real data.

each reported address responds.

²⁵The datasets are mostly public-use surveys and replication packages (e.g., the National Longitudinal Surveys of Youth and the longitudinal studies of the National Center for Education Statistics merged with institutional data in the schooling domains, the Fulton fish market data, the replication package of a social-network field experiment for technology adoption, and the Add Health and Notre Dame NetHealth network studies for teenage smoking), and in more than half of the cases the candidate must be merged from a second source. See Appendix E.3 for details.

8 Variables Search in Other Causal Inference Methods

In this section, we demonstrate how prompting strategies similar to those for the IV discovery can be used to find (i) control variables under which treatments are conditionally independent (i.e., exogenous); (ii) control variables under which parallel trends are likely to hold in difference-in-differences; and (iii) running variables in regression discontinuity designs.

8.1 Conditional Independence

Using the same notation as in Section 2, consider a conditional independence (CI) assumption that assigns a more crucial role to the vector of control variables $X \equiv (X_1, \dots, X_L)$:

Assumption CI. *For any d , $D \perp Y(d)|X$.*

Assumption CI is commonly introduced in causal inference settings, especially when combined with machine learning to estimate nuisance functions; e.g., debiased/double machine learning methods (Chernozhukov et al., 2024). More traditionally, this assumption is closely related to matching and propensity score matching techniques (Heckman et al., 1998). The mean independence version of CI (i.e., $E[Y(d)|D, X] = E[Y(d)|X]$) is relevant to regression methods.

We propose using LLMs to systematically search for X that satisfies a verbal version of CI. The prompt writing is slightly simpler than that for IVs. In particular, we construct prompts that solicit the relationship between X and D (Step 1) and X and $Y(d)$ (Step 2). Therefore, *only* the second-step prompt involves a counterfactual statement. Let L_0 be the number of controls to be found in Step 1 ($L_0 \geq L$). One may want to choose the value of L_0 to be larger than one would normally use for K_0 and leave L unspecified.

Prompt C1 (Search for Control Variables). *you are [agent] who needs to make a [treatment] decision in [scenario]. what factors determine your decision? list [L-0] factors that are quantifiable. explain the answers.*

Prompt C2 (Refine Control Variables). *among the [L-0] factors listed above, choose all factors that directly determine your [outcome], not only indirectly through [treatment]. the chosen factors can still influence your [treatment]. for each factor chosen, explain your reasoning.*

The prompts are constructed to search for confounders and need to be controlled for. Researchers sometimes mistakenly control for “colliders” and/or “mediators” (Pearl, 2000), which are intended to be excluded from the search. Note that Prompts C1–C2 can also be adapted to jointly search for covariates and latent confounders in the IV search. In this case, one can distinguish X from latent confounders by referring to the former as “quantifiable.” Also, one may want to use the phrase “demographic factors” to refer to X , as they are common control variables in many empirical applications.

8.2 Difference in Differences

The difference-in-differences (DiD) method is popular in empirical research, partly due to the simplicity and intuitiveness of its main assumption, namely, the parallel trend assumption (stated below). However, this assumption is not directly testable and typically hard to justify (Ghanem et al., 2022; Rambachan and Roth, 2023). It is believed that conditioning on the right control variables can make this assumption more justifiable, which can motivate the search for such controls.

Assumption PT. $E[\Delta Y(0)|D, X] = E[\Delta Y(0)|X]$ where $\Delta Y(0) \equiv Y_{after}(0) - Y_{before}(0)$.

Assumption PT can be viewed as a mean independence version of CI, where the counterfactual outcome is replaced with the temporal difference of counterfactual (untreated) outcomes before and after the event. Therefore, Prompts C1–C2 can be directly used to search for X that satisfy a verbal version of PT. This can be done by inputting “average temporal changes in [outcome_t] during the time of

no [treatment]” for [outcome] in Prompt C2, where [outcome_t] refers to Y_t for $t \in \{before, after\}$. The example of such prompts is constructed to revisit the classical empirical example, namely, the effects of minimum wage on the fast food industry’s labor markets (Card and Krueger, 1994).

The following prompts are used to produce the results in Table 8.

Prompt C1.1 (Example: Minimum Wage). *you are the policymaker in the department of labor, deciding whether to increase the minimum wage or not and to which state to introduce this minimum wage law. what factors determine your decision? list forty factors that are quantifiable. explain the answers.*

Prompt C2.1 (Example: Minimum Wage). *among the forty factors listed above, choose all factors that directly determine the temporal changes in average wages at fast food restaurants, not only indirectly through the minimum wage law. the chosen factors can still determine your decision of introducing minimum wage law. for each factor chosen, explain your reasoning.*

Table 8 contains the control variables suggested by the LLM, conditional on which the parallel trend is likely to hold, and the LLM’s rationale. On the list, #3, 4, 7, 10, 11 are particularly interesting and #11 seems particularly novel. In the table, the first four rows (#1, 2, 3, 4) are chosen by the LLM from an additional prompt that emphasizes the requirement with respect to $\Delta Y(0)$: “be sure to choose all factors that do not determine the average wage level but only determine the temporal changes in average wages.” Nonetheless, controls that satisfy the mean version of CI with the level, $Y_t(0)$ for $t \in \{before, after\}$, are also valid controls for PT.

#	Control Variables Suggested	Rationale Provided
1	Inflation Rates	Higher inflation can decrease the real income of workers if wages don't adjust accordingly, making a case for introducing a minimum wage to maintain purchasing power.
2	Consumer Price Index (CPI)	A rising CPI without wage increases can reduce workers' real earnings, suggesting a minimum wage could help offset cost of living increases.
3	Job Vacancy Rates	Low vacancies could keep wages down due to ample labor supply; high rates may drive wages up, affecting the urgency and approach to minimum wage legislation.
4	Labor Productivity Growth	This affects temporary wage increases linked to efficiency improvements. Productivity gains can lead to bonus payments or higher wages temporarily but don't define starting wage rates.
5	Employment Growth Rates	Robust job growth in the sector might show the industry can support higher wages, while slow growth suggests caution in wage hikes.
6	Labor Force Participation Rate	High participation may suppress wages due to increased labor supply, supporting the need for a minimum wage to ensure fair earnings.
7	Union Membership Rates	Low unionization in fast food often means weaker wage bargaining, possibly justifying minimum wage laws to improve wage standards.
8	Turnover Rates	High turnover might indicate wage dissatisfaction, pointing to the need for a minimum wage to stabilize the workforce.
9	Corporate Profit Trends	If profits are high but wages are stagnant, it could suggest that the industry can afford to pay more, bolstering the argument for a minimum wage.
10	Economic Diversity Score	Dependency on low-wage sectors like fast food in less economically diverse areas might necessitate a minimum wage to ensure living standards.
11	Percentage of Workforce in Gig Economy	Increased gig work could pressure fast food employers to offer competitive wages, influencing when and how to implement minimum wage laws.

Table 8: Difference-in-Differences for Minimum Wage: Suggested Control Variables

Notes: All controls are discovered and explained by GPT4 from a single run of Prompts C1-C2, adapted to Assumption PT with $L_0 = 40$ and L left unspecified. Among them, the first four rows are factors that are chosen from the additional emphatic prompt: “be sure to choose all factors that do not determine the average wage level but only determine the temporal changes in average wages.” The total running time was less than 1 minute.

8.3 Regression Discontinuity

Regression discontinuity designs (RDDs) are another well-known method for causal inference that closely relates to the IVs method (Lee and Lemieux, 2010).²⁶ The key for this method to work is to find a running variable (i.e., assignment variable) that satisfies the following:

Assumption RD. *There exists a variable R_j and a cutoff r_0 such that $D = 1$ if $R_j \geq r_0$ and $D = 0$ if $R_j < r_0$.*

One can use LLMs to systematically search for running variables $\{R_1, \dots, R_J\}$ for a given D and Y of interest. We provide the example of prompts here. It is worth noting that, unlike in all the previous cases, none of the prompts below involve counterfactual statements. Therefore, if LLMs outperform a traditional search for running variables, it would be due to their automated and comprehensive search behavior. Similarly as above, we only specify initial J_0 and leave J unspecified.

Prompt R1 (Search for Running Variables). *you are [agent] who needs to make a [treatment] decision in [scenario]. what would be the possible criteria based on which your eligibility for [treatment] is determined? provide [J_0] of the most relevant criteria that are (1) quantifiable and (2) have specific cutoffs determining eligibility. explain the answers.*

Prompt R2 (Refine Running Variables). *among the [J_0] criteria listed above, choose all criteria that involve continuous or ordered measures and have precise cutoffs determining eligibility. also report the cutoff value for each criterion from verifiable sources only (ensuring no fabricated or hypothetical numbers are used). explain the answers.*

²⁶For example, the fuzzy RDD estimand can be viewed as the two-stage least squares estimand.

#	Outcome(s) (Country)	Treatment(s)	Suggested Running Variable (Same as the Literature)	Newly Suggested Running Variables (Cutoffs for Eligibility)
1	Spending on schools, test scores (US)	State education aid	Relative average property values (Guryan, 2001)	- Percentage of low-income students (e.g., Equity Multiplier 2023-2024, above 70%) - Mobility rate (e.g., Equity Multiplier, above 25%) - Age (e.g., Transitional Kindergarten (TK) expansion 2023-24, 5th b-day by April 2) - Local Control Funding Formula (LCFF) (California)*
2	College enrollment (US)	Financial aid offer	SAT scores, GPA (Van der Klaauw, 2002)	- Expected family contribution (EFC) (e.g., the Pell Grant 2023-2024: up to \$6,656)
3	Overall insurance coverage (US)	Medicaid eligibility	Age (Card and Shore-Sheppard, 2004)	- Federal Poverty Level (FPL) (e.g., Washington D.C. in 2023: up to 215% for childless adults and 221% for parents/caretaker relatives; equiv. annual incomes up to \$31,347 (household of 1) and \$54,940 (household of 3)) - Household Size (e.g., Modified Adjusted Gross Income (MAGI) rules: expressed as % of FPL, adjusted by 5% FPL disregard)
4	Employment rates (Italy)	Job training program	Attitudinal test score (Battistin and Rettore, 2002) [†]	- Age (e.g., Bonus Giovani hiring incentive, hires Sept. 2024–Dec. 2025: below 35; source: Decree-Law 60/2024) - Salary (e.g., EU Blue Card: at least the average gross annual salary in Italy; source: Legislative Decree 152/2023)
5	Re-employment probability (UK)	Job search assistance, training, education	Age at end of unemployment spell (De Giorgi, 2005)	- Age (e.g., Jobseeker’s Allowance (JSA): 18 or over and under State Pension age; source: GOV.UK) - Minimum Salary (e.g., Skilled Worker visa: at least £41,700 (since July 2025) or going rate for job type, whichever is higher; source: GOV.UK) - Residency Duration (e.g., income-based JSA, 2014–2026 (now abolished): at least 3 months in the Common Travel Area prior to claim; source: SI 2013/3196)

Table 9: Regression Discontinuity: Suggested Candidates for Running Variables

Notes: All running variables are discovered and explained by GPT4 from a single run of Prompts **R1–R2**, adapted to each context with $J_0 = 20$ and J left unspecified. All running variables used in the literature (Column 4) are also found by GPT4, except #4. The total running time for each row was less than 1 minute (even with an automated web search for Prompt **R2**). We verify the cutoffs against official sources and correct or update entries when necessary. The sources indicated are those given by GPT4, except for the corrected entries, for which we indicate the official sources we consulted. *: A formula, not a running variable. †: Not found by GPT4.

Note that when Prompt [R2](#) is run on GPT4, it will engage in a series of automated web searches. The request for cutoff values may lead the LLM to provide hypothetical numbers as possibilities. When one wants to get the actual values from verifiable sources, it is important to explicitly state that, as we do above. We apply Prompts [R1–R2](#) to a range of famous examples in the literature where RDDs are used as empirical strategies. Table [9](#) presents the results obtained by running the prompts, which are adapted to each specific context and country of the empirical example. In most cases, a handful of new possible running variables are suggested by GPT4 with specific cutoffs obtained from web sources. Except for one case (i.e., #4), GPT4 also identifies the running variables used in the literature.

9 Conclusions

This paper proposes the agenda to use LLMs to systematically search for variables in designing causal inference. It merely serves as a starting point, and there are potential next steps that can follow. In constructing prompts for IVs, there are many possible ways for sophistication: First, one can consider using previously known IVs in the literature to guide LLMs to discover new ones. This can be done by adding textual demonstration of how Assumptions [REL–IND](#) are satisfied with known IVs *before* starting the proposed prompts. This approach would evoke few-shot learning in LLMs (as in the LLM judge construction), which can enhance their performances. This approach would also “orthogonalize” the search ([Ludwig and Mullainathan, 2024](#)) to focus on novel IVs. One caveat of this approach is the difficulty of masking the discovery task. Second, the elaborated search can be directed toward finding IVs that are more policy-relevant ([Imbens and Angrist, 1994](#); [Heckman and Vytlacil, 2005](#)) by specifying targeted policies in the prompt. Third, the discovery results in Part I are from single sessions. Section [7](#) shows one way of aggregating across sessions (i.e., grouping candidates into instrument families) to account for and potentially leverage the stochastic nature of LLMs’ responses. Aggregating outputs across different LLMs (e.g.,

GPT and Claude) would also be beneficial.²⁷

One challenge of our approach—shared with other hypothesis-generation procedures—is that the performance metric is difficult to define due to the lack of ground truth for valid IVs. In fact, this is the very reason we propose using LLMs in the first place: for any IVs found by human researchers or the machine, there are only more compelling narratives or less compelling ones. We partly address this challenge in the evaluation pipeline, through an expert survey in Section 6 and an LLM judge calibrated to the survey in Section 7. In later stages, when data eventually come into play, over-identification tests can potentially be a fruitful framework for the evaluation of LLMs. More broadly, evaluating hypothesis-generating systems remains an interesting open question. We still believe the paper’s proposal serves as a thought-provoking starting point. It would be interesting to apply the proposed variable-search approach to other empirical settings and causal inference methods.

²⁷Similar to running multiple sessions within an LLM, responses of multiple LLMs can be correlated and potentially biased (Kim et al., 2025). A suitable aggregation strategy akin to that discussed in Section 7 can resolve the issue of correlation, and the expert survey and the LLM judge may be able to detect possible bias.

A Three-Step Prompts with System Messages

We present three-step prompts, where each step corresponds to one of **REL**, **EX** and **IND**. This version includes *system messages* that establish the scenario (including institutional details) and role of interest as context information and response formats before initiating the main prompts.

Prompt A0 (System Messages). *you are [agent] who needs to make a [treatment] decision in [scenario]. you will be given questions on your decisions and related outcomes. in answering the questions, always provide explanations for your answers. each answer should be within five words and each explanation should be within twenty words. do you understand your role, the scenario you are in, and the instructions given to you?*

Prompt A1 (Search for Relevance). *what are factors that can determine your decision? list [K_0] factors that are quantifiable. explain the answers.*

Prompt A2 (Search for Exclusion). *among [K_0] factors listed above, what are factors that affect your [outcome] only through [treatment]? explain the answers.*

Prompt A3 (Search for Independence). *among the factors listed above, choose [K] factors that are most likely to be unassociated with [confounders], which determine your [outcome]. the chosen factors can still influence your [treatment]. for each factor chosen, explain your reasoning.*

B Alternative Prompts with Covariate Search

Instead of using user-specified covariates in Prompt [2_x](#), an alternative way is to search for covariates in Step 1. Prompt [1_x](#) below is designed to jointly search for (Z_k, X) that satisfy EX and [REL2](#):

Assumption REL2. (i) The distribution of D given $(Z_k, X) = (z_k, x)$ is a nontrivial function of (z_k, x) and (ii) the distribution of $Y(d)$ given $X = x$ is a nontrivial function of x .

Prompt 1_x (Search for IVs and Covariates). *you are [agent] who needs to make a [treatment] decision in [setting]. what are factors that can determine your decision but that do not directly affect your [outcome], except through [treatment] (that is, factors that affect your [outcome] only through [treatment])? list [K_0] factors. also, what are your characteristics that directly influence [treatment] and directly influence [outcome] (not just through [treatment])? list [L_0] characteristics. explain the answers.*

Instead of running Prompt [1](#) in Step 1, one can run Prompt [1_x](#) to find an initial set of covariates and select a subset among them at her discretion to run Prompt [2_x](#) in Step 2.

C Prompts Used in Applications (Section [4](#))

C.1 Production Function Estimation

We choose $K_0 = 40$ and let GPT4 choose K ; we explicitly request separate lists for market factors and firm and manager factors. Note that in Prompt [2_x.2.1](#), we use a loose description of covariates (unlike in Prompt [2_x.1](#) in Section [4.1](#) where we assign specific values).

Prompt 1.2.1 (Example: Production Functions). *you are a manager at a manufacturing firm. you need to make a decision on how much*

labor and capital inputs to use to produce outputs. what would be factors (factors of markets and economy and factors of yourself) that can determine your decision but that do not directly affect your output productions, except through the input choices (that is, that affect your firm's outputs only through inputs)? list forty factors that are quantifiable, twenty for market factors and twenty for managerial factors. explain the answers.

Prompt 2_x.2.1 (Example: Production Functions). *suppose you are a manager at a firm with specific level of capital intensity and specific scale of operations, which has a specific market share in a specific industry. among the forty factors listed above, choose all factors that are not influenced by productivity shocks of your firm, which determine outputs. create separate lists for market factors and managerial factors. for each factor chosen, explain your reasoning.*

C.2 Peer Effects

C.2.1 Effects of Peer Farmers on New Technology Adoption

It is worth noting that, in this example, the role-playing is done from the peer's perspective, rather than from the perspective of the individual whose outcome is of concern.

Prompt 1.3.1 (Example: Peer Effects on Technology Adoption). *you are a farmer in a village in rural india. you want to influence your peer farmers in the same village to introduce a new farming technologies that you introduced. what would be factors (factors of farming and village, and factors of yourself) that can determine your influence on peers but that do not directly*

affect your peers' technology adoption decisions, except through your influence (that is, that affect your peers' decisions only through your influence)? list forty factors that are quantifiable. explain your answer.

Prompt 2_x.3.1 (Example: Peer Effects on Technology Adoption). *suppose you are a 40 year old male farmer of a specific crop in the village in rural india. among the factors listed above, which factors are not influenced by factors (e.g., similar background and preferences) that brought you and your peers in the same neighborhood and social network from the first place? for each factor chosen, explain your reasoning.*

C.2.2 Effects of Peer Teenagers on Smoking Behavior

Again, in this example, the role-playing is done from the peer's perspective.

Prompt 1.3.2 (Example: Peer Effects on Smoking). *you are a teenager in indonesia who smokes. you want to influence your peers in your social media network to smoke. what would be factors (factors of social media, your school and region, and factors of yourself) that can determine your influence on peers but that do not directly affect your peers' smoking decisions, except through your influence (that is, that affect your peers' decisions only through your influence)? list forty factors that are quantifiable. explain your answer.*

Prompt 2_x.3.2 (Example: Peer Effects on Smoking). *suppose you are a teenage boy in urban indonesia who goes to high school and is from middle-income family. among the factors listed above, which factors are not influenced by factors (e.g., similar background*

and preferences) that brought you and your peers in the same social network from the first place? for each factor chosen, explain your reasoning.

D Additional Tables for Discovered IV Candidates

(Section 4)

#	IVs Suggested	Rationale Provided
1	Interest Rates	Influence the cost of borrowing, affecting decisions on how much capital to acquire or lease for production without altering current productivity levels.
2	Exchange Rates	Affect the price of imported inputs, which can change input costs and investment decisions in foreign capital, again without directly impacting the productivity of existing inputs.
3	Tariffs and Trade Policies	Alter the cost structures for imported and exported goods, impacting decisions on where to source inputs or sell outputs, but do not affect the productivity of the inputs themselves.
4	Regulatory Changes	Can necessitate changes in production processes or input materials, which may affect costs and sourcing decisions without directly impacting the productivity of the inputs once deployed.
5	Transportation Costs	Affect the total cost of inputs and can influence decisions regarding sourcing and logistics. These costs don't directly alter how effectively inputs are converted into outputs.
6	Real Estate Prices	Influence decisions about expanding or relocating production facilities, affecting capital expenditure and operational scale without altering current production efficiency.
7	Environmental Regulations	Stricter environmental regulations may force a firm to switch to greener, possibly more expensive inputs or production technologies, affecting input choices and costs but not directly modifying the productivity of any specific input set.
8	Patents and Intellectual Property Rights	Governed by legal systems and not directly influenced by a firm's internal productivity shocks.
9	Political Stability	Affect market predictability and investment decisions, impacting where and how much to invest in capital and labor, without changing how these inputs produce once acquired.
10	Global Economic Conditions	Influence market opportunities and risks associated with foreign investments, guiding input acquisition strategies without changing input productivity.

Table A1: (a) Production Functions: Suggested Candidates for IVs (Market Factors)

Notes: IV candidates for inputs are presented. All IV candidates are discovered and explained by GPT4 from a single run of Prompts 1.2.1 and 2_x.2.1 with $K_0 = 40$ and K left unspecified. Panel (a) is categorized by GPT4 to be market-related factors, and Panel (b) to be managerial factors. The rationale given by LLM can be elaborated further by requesting it in the same session. The total running time was less than 1 minute.

#	IVs Suggested	Rationale Provided
11	Capital Availability	Determines its ability to invest in both labor and capital inputs, shaping the scale of operations and types of technology employed, without directly changing how productive these inputs are
12	Company's Risk Tolerance	Can decide the extent to which a firm is willing to invest in new, potentially more efficient, but riskier technologies or markets, affecting input decisions rather than the productivity of current inputs
13	Strategic Objectives	Long-term strategic objectives may dictate prioritizing certain types of inputs or production scales, influencing the firm's approach to markets and technology investments without affecting current input productivity.
14	Financial Health of the Company	The overall financial stability can limit or expand the firm's ability to procure and utilize inputs optimally, shaping how inputs are managed and financed rather than directly influencing their productivity
15	Compliance and Legal Considerations:	Driven by external legal requirements and internal ethics, not by short-term productivity
16	Corporate Social Responsibility (CSR) Initiatives	Strategic decisions about CSR are influenced by long-term planning and brand image considerations.

Table A1: (b) Production Functions: Suggested Candidates for IVs (Managerial Factors)

Notes: IV candidates for production inputs are presented. All IV candidates are discovered and explained by GPT4 from a single run of Prompts 1.2.1 and 2_x.2.1 with $K_0 = 40$ and K left unspecified. Panel (a) is categorized by GPT4 to be market-related factors, and Panel (b) to be managerial. The rationale given by LLM can be elaborated further by requesting it in the same session. The total running time was less than 1 minute.

#	IVs Suggested	Rationale Provided
1	Number of Training Sessions Attended	This reflects an individual’s initiative to seek additional education and may vary widely among farmers with similar backgrounds due to personal choices, availability of time, or perception of the value of formal education
2	Number of Demonstrations Conducted	This is influenced by an individual’s willingness to share knowledge and the resources they are willing or able to allocate for such activities, which might not be uniformly distributed even among neighbors
3	Increase in Yield After Technology Adoption	This factor depends significantly on how effectively an individual implements and adapts new technologies to their specific farming practices, which can differ greatly even among farmers growing the same crops
4	Access to Credit	Individual financial management skills, credit history, and relationships with lending institutions can influence this factor, which isn’t necessarily shared among all peers in a community
5	Participation in Local Decision Making Bodies	Involvement in local governance or committees often depends on personal ambition, political skills, and community perception of an individual’s leadership qualities, which can vary independently of shared initial factors
6	Awards or Recognitions Received	Achieving recognition for farming practices or community service is heavily dependent on individual effort and excellence and is not uniformly influenced by community-shared characteristics
7	Communications Skills Assessment	Communication effectiveness can be developed through personal experience, education, and innate ability, which vary among individuals regardless of their initial common background
8	Number of Articles or Case Studies Published	This is influenced by an individual’s ability to write, research, and engage with broader audiences, skills that are not common to all individuals in a network
9	Engagement in Social Media or Technology Platforms	The level of engagement in modern communication and technology platforms depends on personal interest in technology, availability of resources, and individual attitudes toward innovation
10	Success Stories Documented and Shared	The ability and willingness to document and share success stories can be a personal choice influenced by an individual’s perspective on self-promotion and communication

Table A2: Peer Effects on Technology Adoption: Suggested Candidates for IVs

Notes: IV candidates for peers’ decisions are presented. Given the perspective in the prompts, the variables should be understood as factors of *peers*. All IV candidates are discovered and explained by GPT4 from a single run of Prompts 1.3.1 and 2_x.3.1 with $K_0 = 40$ and K left unspecified. The rationale given by LLM can be elaborated further by requesting it in the same session. The total running time was less than 1 minute.

#	IVs Suggested	Rationale Provided
1	Engagement Rate	While the number of followers might be initially influenced by shared backgrounds, the engagement rate depends more on the content quality and how it resonates with the audience at any given time, rather than the reasons why the audience initially formed
2	Frequency of Posts	This is largely a personal choice and strategic decision, independent of peer background. It reflects personal commitment to the platform rather than shared interests or backgrounds
3	Content Reach	This metric depends on the algorithm and how engaging the content is, which can transcend the initial reasons for network formation
4	Number of Platforms Used	Choosing to diversify presence across various platforms is a strategic decision that doesn't necessarily correlate with shared backgrounds or interests
5	School Policies on Smoking	These are institutional decisions that apply to all students, regardless of the factors that brought them together
6	Local Regulations on Tobacco Advertising	Like school policies, these are determined by local government and affect all residents uniformly, irrespective of social network formations
7	Access to Smartphones Among Peers	This might vary widely even within similar socio-economic backgrounds due to individual family decisions or priorities
8	Visibility at School Events	Active participation and visibility in school events are personal choices and efforts that go beyond shared backgrounds, reflecting individual initiative
9	History of Disciplinary Actions at School	This is generally a result of personal behavior and choices rather than group influence
10	Academic Performance	Although there could be a correlation with socio-economic status, individual effort and capability play significant roles, making this somewhat independent of why peers might group together initially
11	Extracurricular Leadership Roles	Holding leadership positions is often based on personal qualities, skills, and choices rather than the shared preferences and backgrounds that might define a social network initially

Table A3: Peer Effects on Smoking: Suggested Candidates for IVs

Notes: IV candidates for peers' decisions are presented. Given the perspective in the prompts, the variables should be understood as factors of *peers*. All IV candidates are discovered and explained by GPT4 from a single run of Prompts 1.3.2 and 2.x.3.2 with $K_0 = 40$ and K left unspecified. The first four rows concern social media factors; the next three rows concern school and regional factors; and the last four rows concern personal factors. The rationale given by LLM can be elaborated further by requesting it in the same session. The total running time was less than 1 minute.

E Details of LLM Judge Evaluation (Section 7)

E.1 Prompts for the Alternative Strategies

We present the prompts of the five strategies compared in Section 7 for the returns to college example (Section 4.1.1); the prompts for the other domains are constructed in the same way. In every strategy, the LLM is asked for twenty IV candidates in the final step.

The two-step role-playing strategy (1) uses Prompts 1.1 and 2_x.1, except that Prompt 2_x.1 asks to “choose 20 factors” instead of “choose all factors.” The one-step role-playing strategy (2) merges the two prompts into Prompt E1.

Prompt E1 (One-Step Role-Play). *you are a high school graduate with family income \$10K per year, who is asian female from california, whose parents have college education, who is catholic. you need to make a college attendance decision. what would be factors (factors of schools and factors of yourself) that can determine your decision but that do not directly affect your future earnings, except through college attendance (that is, that affect your earnings only through college attendance), and that are not associated with your innate ability and personality and school quality, which determine earnings? choose 20 factors that are quantifiable. create separate lists for school factors and factors of yourself. for each factor chosen, explain your reasoning.*

The direct strategies (3)–(5) start with Prompt E2, which gives the definition of a valid IV. The two-step direct strategy (3) then runs Prompts E3a and E3b, which mirror Prompts 1.1 and 2_x.1; the one-step direct strategy (4) runs Prompt E4 instead; and the novelty-encouraged strategy (5) runs Prompt E4 with the sentence in Prompt E5 appended. The covariates in these prompts are the characteristics of the student in Prompt 2_x.1.

Prompt E2 (Definition of IV). *here is the definition of instrumental variables (IV): IV is used to estimate causal effect of a treatment on outcome when treatment is endogenous in the sense that there are unobserved confounders that simultaneously affect the treatment and outcome. IV is a variable that is correlated with the treatment and affects the outcome only through the treatment and IV is independent of the unobserved confounders. Do you understand the definition of IV?*

Prompt E3a (Two-Step Direct, Step 1). *now, suppose the treatment is college attendance, the outcome is wages and the unobserved confounders are students' ability, personality and school quality. suggest me 40 candidate variables that are correlated with the treatment and that affect the outcome only through the treatment. the variables should be quantifiable. explain your answers.*

Prompt E3b (Two-Step Direct, Step 2). *among the 40 variables listed above, choose 20 variables that are also independent of the unobserved confounders, so that they would be valid instrumental variables conditional on family income, race/ethnicity, gender, state of residence, parental education, religion. for each variable chosen, explain your reasoning.*

Prompt E4 (One-Step Direct). *now, suppose the treatment is college attendance, the outcome is wages and the unobserved confounders are students' ability, personality and school quality. suggest me 20 instrumental variables that would be valid conditional on family income, race/ethnicity, gender, state of residence, parental education, religion*

Prompt E5 (Emphasis on Novelty). *as much as possible, suggest instrumental variables that have not been used in the existing empirical literature on this question (that is, novel instruments), while making sure that they still satisfy the definition of a valid instrument given above.*

In every strategy, the candidates of a run are then converted into standardized *IV cards* (i.e., the variable name and one- or two-sentence rationales for **REL**, **EX** and **IND**), which are written by an independent LLM from the output of the discovery run (Prompt **E6**, run with GPT5 in batches of five candidates). This way, the format of the prompt cannot reveal which strategy produced the candidate. As in Section 3, each bracketed term is filled in from the domain and the run. In the role-playing strategies, the prompt additionally quotes the role-playing prompt that specifies the covariates (Prompt **2_x.1** in the two-step strategy and Prompt **E1** in the one-step strategy) and asks to extract the control variables mentioned in it; the judge does not use this field.

Prompt E6 (IV Card). *You are an econometrician. For each candidate instrumental variable listed below, produce a structured "IV card" with exactly the following fields.*

Context:

- Treatment (D): [treatment]*
- Outcome (Y): [outcome]*
- Unobserved confounders (U): [confounders]*

Candidate instruments from the discovery procedure:

[list of the candidates of the run, each with the rationale given by the LLM]

For EACH instrument, output a JSON object with these fields:

"name": short instrument label (string),

"relevance": 1-2 sentences on the mechanism $Z \rightarrow D$ (string),

"exclusion": 1-2 sentences on why Z does NOT directly affect Y except through D (string),

"independence": 1-2 sentences on why Z is independent of U conditional on controls X (string),

"controls": candidate control variables X (list of strings)

Output a JSON array of objects, one per instrument.

Output ONLY valid JSON, no markdown fences, no commentary.

E.2 The LLM Judge Construction

The judge (GPT5) receives the system message in Prompt E7 and the user message in Prompt E8, and returns the two scores as a JSON object. The setting lines of Prompt E8 are shown for the returns to college example; for the two peer-effects domains, they include the caution that “The variable should be understood as factors of peers (not factors of individual receiving peer effects).” The expert-scored examples are all IVs of the candidate’s domain in the survey of Section 6 (seven to sixteen per domain), listed in increasing order of denoised validity, each with the rationale shown to the experts and its denoised scores rounded to integers. The message asks the judge to return the two integer scores in a fixed format.

In the evaluation stage, each of the 70 IVs in the expert survey is scored in turn as a hold-out IV that is removed from the few-shot examples. When each hold-out IV is scored, the judge is presented with the rationale shown to the experts. When evaluated (Figure 2), we find that the rank correlation between the judge’s scores and the denoised expert scores is 0.64 for validity and 0.78 for non-novelty. In leave-one-out cross-validation, the mean absolute error of the calibrated judge scores is 0.27 for validity and 0.30 for non-novelty on the 1–5 scale, compared with 0.33 and 0.44 for a naive predictor that assigns every IV the average score of its domain.

Prompt E7 (LLM Judge: System Message). *You are an expert academic*

economist evaluating candidate instrumental variables (IVs). Here is the definition of a valid instrumental variable (IV): An IV is used to estimate the causal effect of a treatment on an outcome when the treatment is endogenous in the sense that there are unobserved confounders that simultaneously affect the treatment and the outcome. An IV is a variable that satisfies the following conditions: (i) it is correlated with the treatment; (ii) it affects the outcome only through the treatment; and (iii) it is independent of the unobserved confounders.

You will be shown an IV candidate with its rationale, in the context of a specific econometric setting.

Your task is to provide two ratings:

You will rate one candidate IV at a time, in the econometric setting described in the next message. The candidate was proposed during an instrument-discovery exercise; its rationale is shown for reference when available. For each IV, please provide:

A validity grade (1 = definitely not valid, up to 5 = definitely valid)

A confidence rating on whether it already exists in the literature (1 = low confidence it exists or equivalently 1 = most novel, up to 5 = high confidence it exists or equivalently 5 = least novel)

Please focus primarily on the variable name; use the rationale text to inform your judgment of the instrument's plausibility, but grade the instrument, not the quality of the write-up. If no rationale is shown, judge from the variable name alone.

Output ONLY a JSON object with two integer fields:
 {"validity": <int 1-5>, "nonnovelty": <int 1-5>}
 IMPORTANT: Use the full 1-5 scale. Do not compress your ratings to a narrow range. A score of 1 means definitely not valid / most novel, and 5 means definitely valid / high confidence it exists in the literature.

Prompt E8 (LLM Judge: User Message). *Setting:* Returns to College
Setup: High-school graduate deciding whether to attend college.
Endogeneity: Unobserved ability & school quality jointly influence attendance and earnings.
 --- EXPERT-RATED EXAMPLES (from this same setting, ordered from lowest to highest validity) ---
IV suggested: [survey IV name]
Rationale Provided: [survey IV rationale]
Expected rating: {"validity": [denoised validity],
 "nonnovelty": [denoised non-novelty]}
 [one such block for each survey IV of the domain]
 --- IV TO EVALUATE ---
IV suggested: [candidate name]
Rationale Provided: Relevance: [relevance] Exclusion:
 [exclusion] Independence: [independence]
 Provide your rating as a JSON object:

E.3 Additional Tables

Tables A4–A7 list, for each strategy, the instrument families with non-novelty score $k = 1$ or 2 whose average calibrated validity is at least the benchmark of their domain (the LLM-judge analog of Table 5). No candidate from the one-step direct strategy met this criterion, so *no*

table is reported for it. Tables A8–A13 list, for each domain, the datasets found for the novel and valid candidates in Section 7.4.

Domain [†] (Benchmark) ^{††}	Novel IVs Suggested ^{†††}
1 (2.59)	Campus Safety Statistics (+0.13)
2 (2.28)	School Start Times (+0.25)
	Access to Online Learning (+0.06)
	Library Book Count (+0.06)
	School Wi-Fi Quality (+0.06)
	Access to School Lunch Programs (+0.06)
	Mental Health Support (+0.06)
	Access to Internships (+0.06)
	Art/Music Program Availability (+0.06)
	School Sports Team Success (+0.06)
	Extracurricular Offerings (+0.06)
	Availability of Electives (+0.06)
	Community Support (+0.06)
	Mentorship Opportunities (+0.06)
	Access to Tutoring Services (+0.06)
	Number of Guidance Counselor Meetings (+0.06)
	Annual Field Trips (+0.06)
	School Anti-Bullying Policies (+0.06)
	Participation in College Fairs (+0.06)
	School Events/Activities (+0.06)
	Exposure to Role Models (+0.06)
	Cultural Values on Education (+0.06)
	School Discipline Policies (+0.06)
	Parent Satisfaction Surveys (+0.06)
5 (2.53)	Collaborations with Experts (+0.20)
	Media Presence (+0.20)
	Number of Demonstration Plots (+0.20)
	Number of Training Sessions Conducted (+0.20)
	Number of Technology Demonstrations Conducted (+0.20)
	Positive Testimonials Count (+0.20)
	Successful Case Studies Shared (+0.20)
	Workshops Hosted (+0.20)
	Public Speaking Engagements (+0.20)

Table A4: (a) LLM Judge: IVs with $k = 1$ or 2 Outperforming Benchmark: Two-step role-play (Domains 1, 2, 5) ($k = 1$: most novel; $k = 5$: least novel)

[†]Domain ID: 1. Returns to College Attendance; 2. Returns to Years of Education; 3. Demand Estimation; 4. Production Functions; 5. Peer Effects (technology adoption); 6. Peer Effects (teenage smoking). ^{††}The benchmark is the average calibrated validity of the strategy’s candidates with non-novelty ≥ 4 in the domain, in parenthesis. ^{†††}An instrument family is listed if the average calibrated validity of its novel candidates ($k = 1$ or 2) across runs is at least the benchmark; the improvement over the benchmark is in parenthesis. Domains without qualifying candidates are omitted. Candidate names are as generated by the LLM.

Domain [†] (Benchmark) ^{††}	Novel IVs Suggested ^{†††}
6 (2.32)	Content Reach (+0.41) Social Media Advertising (+0.41) Visual Appeal (+0.41) Frequency of Posts (+0.41) Platform Diversity (+0.41) Story Views (+0.41) Influence Score (+0.41) Content Quality (+0.41) Engagement Rate (+0.41) Sentiment of Comments (+0.41) Hashtag Usage (+0.36) Content Variety (+0.34) Number of Followers (+0.24) Use of Influencers/Endorsements (+0.11) School Popularity (+0.02) School Social Group Reach (+0.02) Cultural Influence Index (+0.02) Peer Networks (+0.02) Communication Skills (+0.02) Language Use (+0.02) Leadership Role Count (+0.02) Lifestyle Display (+0.02) Regional Trends (+0.02) Local Celebrity Status (+0.02) Risk-Taking Behavior (+0.02) Participation in School Activities (+0.02) Resilience to Criticism (+0.02)

Table A4: (b) LLM Judge: IVs with $k = 1$ or 2 Outperforming Benchmark: Two-step role-play (Domain 6) ($k = 1$: most novel; $k = 5$: least novel)

[†]Domain ID: 1. Returns to College Attendance; 2. Returns to Years of Education; 3. Demand Estimation; 4. Production Functions; 5. Peer Effects (technology adoption); 6. Peer Effects (teenage smoking). ^{††}The benchmark is the average calibrated validity of the strategy’s candidates with non-novelty ≥ 4 in the domain, in parenthesis. ^{†††}An instrument family is listed if the average calibrated validity of its novel candidates ($k = 1$ or 2) across runs is at least the benchmark; the improvement over the benchmark is in parenthesis. Domains without qualifying candidates are omitted. Candidate names are as generated by the LLM.

Domain [†] (Benchmark) ^{††}	Novel IVs Suggested ^{†††}
1 (2.71)	On-Campus Housing Availability (+0.02) Campus Safety (+0.02) Religious Services (+0.02)
5 (2.61)	Willingness to Share Knowledge (+0.12) Record Keeping of Farming Practices (+0.12) Success Stories Shared (+0.12) Training or Workshops Conducted (+0.12) Number of Demonstrations Held (+0.12) Number of Farmers Mentored (+0.12) Health and Nutrition Advocacy (+0.12) NumberOfWorkshopsConducted (+0.12)
6 (2.38)	Type of Content (+0.34) Visual Creativity Score (+0.34) Content Authenticity (+0.34) Engagement Rate (+0.34) Use of Story Features (+0.34) Engagement Tools (+0.34) Number of Social Media Platforms Used (+0.34) Hashtag Usage (+0.34) Frequency of Posts (+0.32) Content Quality (+0.30) Follower Count (+0.29) Influence Score (+0.25) Language Use (+0.09) Peer Mentions (+0.03) Mutual Friends' Support (+0.01)

Table A5: LLM Judge: IVs with $k = 1$ or 2 Outperforming Benchmark: One-step role-play ($k = 1$: most novel; $k = 5$: least novel)

[†]Domain ID: 1. Returns to College Attendance; 2. Returns to Years of Education; 3. Demand Estimation; 4. Production Functions; 5. Peer Effects (technology adoption); 6. Peer Effects (teenage smoking). ^{††}The benchmark is the average calibrated validity of the strategy's candidates with non-novelty ≥ 4 in the domain, in parenthesis. ^{†††}An instrument family is listed if the average calibrated validity of its novel candidates ($k = 1$ or 2) across runs is at least the benchmark; the improvement over the benchmark is in parenthesis. Domains without qualifying candidates are omitted. Candidate names are as generated by the LLM.

Domain [†] (Benchmark) ^{††}	Novel IVs Suggested ^{†††}
5 (2.67)	CharitableActivities (+0.06) Attendance at Local Government Meetings (+0.06)

Table A6: LLM Judge: IVs with $k = 1$ or 2 Outperforming Benchmark: Two-step direct ($k = 1$: most novel; $k = 5$: least novel)

[†]Domain ID: 1. Returns to College Attendance; 2. Returns to Years of Education; 3. Demand Estimation; 4. Production Functions; 5. Peer Effects (technology adoption); 6. Peer Effects (teenage smoking). ^{††}The benchmark is the average calibrated validity of the strategy’s candidates with non-novelty ≥ 4 in the domain, in parenthesis. ^{†††}An instrument family is listed if the average calibrated validity of its novel candidates ($k = 1$ or 2) across runs is at least the benchmark; the improvement over the benchmark is in parenthesis. Domains without qualifying candidates are omitted. Candidate names are as generated by the LLM.

Domain [†] (Benchmark) ^{††}	Novel IVs Suggested ^{†††}
6 (2.73)	Local Cybersecurity Incidents (+0.19) Shared Viral Content Access (+0.00) Time Zone Engagement (+0.00) Local Social Media Workshops (+0.00)

Table A7: LLM Judge: IVs with $k = 1$ or 2 Outperforming Benchmark: One-step direct (novelty-encouraged) ($k = 1$: most novel; $k = 5$: least novel)

[†]Domain ID: 1. Returns to College Attendance; 2. Returns to Years of Education; 3. Demand Estimation; 4. Production Functions; 5. Peer Effects (technology adoption); 6. Peer Effects (teenage smoking). ^{††}The benchmark is the average calibrated validity of the strategy’s candidates with non-novelty ≥ 4 in the domain, in parenthesis. ^{†††}An instrument family is listed if the average calibrated validity of its novel candidates ($k = 1$ or 2) across runs is at least the benchmark; the improvement over the benchmark is in parenthesis. Domains without qualifying candidates are omitted. Candidate names are as generated by the LLM.

Table A8: Datasets for candidate IV families: Returns to College Attendance (Domain 1).

Datasets include these variables: treatment (D) = College attendance (whether the person attended college); outcome (Y) = Wages / earnings in adulthood; covariates (X) = family income, race/ethnicity, gender, state of residence, parental education, religion (where available). Unobserved confounders: Students' ability, personality and school quality.

#	Instrument family (members)	V	N	Data source (link)	How to get it
1	Campus Safety Statistics (+2: Campus Safety; Safety Index)	3	2	ELS:2002	Free public-use download via NCES Online Codebook/DataLab; create extract and download. Restricted-use license only needed for sensitive geography or PETS files.
<i>same family, other data source</i>				NLSY97	NLSY97 public-use via NLS Investigator (free); apply for restricted-use geocode/college-ID file from BLS to enable merging. Campus Safety and Security (Clery) institution-year files are free to download from OPE's Campus Safety site.
2	Campus Housing Options (+1: Housing Options)	3	2	NLSY97	NLSY97 public-use via NLS Investigator (free). To link colleges, apply for NLSY97 restricted-use transcript/geocode license via ResearchDataGov; IPEDS IC/IC_AY files are free downloads from the IPEDS Data Center.
<i>same family, other data source</i>				NLSY79	NLSY79 main data are free (public-use); geocode/college identifiers require a restricted-use data application to BLS. IPEDS IC/COST files are free downloads from the NCES IPEDS Data Center.

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#	Instrument family (members)	V	N	Data source (link)	How to get it
3	Meal Plan Costs	3	2	NLSY97	NLSY97 main and PSTRAN: free download via NLS Investigator; IPEDS: free via IPEDS Data Center; NLSY97 IPEDS UNITIDs needed for the merge are on the restricted-use Geocode files (apply for a license).
4	Weather and Climate	3	2	NLSY79	NLSY79 public-use: free via NLS Investigator. Restricted geocodes (state/county/ZIP/tract and college identifiers): apply through ResearchDataGov SAP; use at BLS or FSRDCs. NOAA nClimGrid-Daily: free public download from NCEI.

Notes: 7 candidate IV names from the two-step role-play strategy (pooled across runs; raw LLM-judge non-novelty rating at most 2 and validity rating at least 3 on at least one card), grouped into 4 instrument families by the judge pipeline’s embedding clustering; further family members are listed in parentheses. V (N) is the maximum validity (minimum non-novelty) rating over the family’s novel-and-valid cards. One row per family and distinct data source; a family whose members found several data sources has several rows. NA = no dataset found for any member.

Table A9: Datasets for candidate IV families: Returns to Years of Education (Domain 2).

Datasets include these variables: treatment (D) = Years of completed education; outcome (Y) = Wages / earnings in adulthood; covariates (X) = family income, race/ethnicity, gender, state of residence, parental education, religion (where available). Unobserved confounders: Innate ability, personality, school quality and other determinants of earnings.

#	Instrument family (members)	V	N	Data source (link)	How to get it
1	School Start Times	3	2	Add Health	Public-use download (registration) for core files; School Administrator files and full linking typically require restricted-use data agreement via Add Health/CPC.

Notes: 1 candidate IV name from the two-step role-play strategy (pooled across runs; raw LLM-judge non-novelty rating at most 2 and validity rating at least 3 on at least one card), grouped into 1 instrument family by the judge pipeline’s embedding clustering; further family members are listed in parentheses. V (N) is the maximum validity (minimum non-novelty) rating over the family’s novel-and-valid cards. One row per family and distinct data source; a family whose members found several data sources has several rows. NA = no dataset found for any member.

Table A10: Datasets for candidate IV families: Demand Estimation in a Fish Market (Domain 3).

Datasets include these variables: treatment (D) = Price of fish (endogenous regressor); outcome (Y) = Quantity of fish demanded / sold; covariates (X) = none specified (where available). Unobserved confounders: Fish market conditions and customer characteristics that shift demand.

#	Instrument family (members)	V	N	Data source (link)	How to get it
1	Waste Management Costs (+2: Waste Disposal Costs; Waste Management Fees)	3	2	catalog.data.metro.tokyo.lg.jp/dataset/t00001...	Free CSV download (Tokyo Metropolitan Government Open Data). Waste-fee data: public pages and CSV via Tokyo 23 Wards Clean Association/wards.
2	Technology Costs (+2: Technology Investment; Technological Investment)	3	2	doi:10.7910/dvn/bxsyl	Free download from Harvard Dataverse (open access).
	<i>same family, other data source</i>			data.norge.no/en/datasets/e5d67089-38d5-4c20-...	Free open data (CSV). Landing page above; documentation with variable list and definitions is on Fiskeridirektoratet; last 12 months of prices are excluded by policy.
3	Depreciation (+2: Depreciation of Equipment; Equipment Depreciation)	3	2	fisheries.noaa.gov/west-coast/science-data	Restricted/confidential microdata; aggregated data public. For research access, contact the NWFSC EDC team listed on the page (data protected under Magnuson–Stevens Act).
	<i>same family, other data source</i>			data.norge.no/en/datasets/e5d67089-38d5-4c20-...	Open CSV download under NLOD; documentation PDF on variables. Instrument comes from a separate, restricted research dataset (apply to Fiskeridirektoratet).
4	Equipment Maintenance (+1: Equipment Maintenance Costs)	3	2	mcc.icpsr.umich.edu/evaluations/catalog/17684	No public files shown on landing page; request via MCC Evidence Platform/ICPSR contact.

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#	Instrument family (members)	V	N	Data source (link)	How to get it
5	Utilities Expenses (+1: Utility Costs)	3	2	franceagrimer.fr/chiffre-et-analyses-economiq...	FranceAgriMer data: free download (weekly/monthly/annual aggregates; Excel/PDF/ZIP). ENTSO-E prices: free via Transparency Platform (CSV/API; free account/FTP).
<i>same family, other data source</i>				spec.api.metro.tokyo.lg.jp/spec/t000013d00000...	Free public data. Fish prices/quantities downloadable via Tokyo Metropolitan Open Data CSV/API; JEPX market data downloadable from JEPX Market Data pages (select date/range, click Data Download).
6	Insurance Costs	3	2	Fulton Fish Market (Angrist-Graddy-Imbens / G...	Free download from Angrist Data Archive; instrument series from NY Department of Insurance 2007 Annual Report PDF (free).
7	Security Costs	3	2	reports.pfex.org/customreport.asp	Free web query; CSV download from the PFEX Custom Report tool.
8	Packaging Costs	3	2	Fulton Fish Market (Angrist-Graddy-Imbens / G...	Free download via R CRAN package fixest; BLS PPI series are free via the PPI Databases page (Series Report/text files).
9	Quality Control Costs	3	2	NA	NA
10	Legal Fees	3	2	NA	NA
11	Supplier Reliability	4	2	NA	NA
12	Training Programs	3	2	NA	NA

Notes: 20 candidate IV names from the two-step role-play strategy (pooled across runs; raw LLM-judge non-novelty rating at most 2 and validity rating at least 3 on at least one card), grouped into 12 instrument families by the judge pipeline’s embedding clustering; further family members are listed in parentheses. V (N) is the maximum validity (minimum non-novelty) rating over the family’s novel-and-valid cards. One row per family and distinct data source; a family whose members found several data sources has several rows. NA = no dataset found for any member.

Table A11: Datasets for candidate IV families: Production Function Estimation (Domain 4).

Datasets include these variables: treatment (D) = Labor and capital input choices; outcome (Y) = Firm output (production, revenue or value added); covariates (X) = capital intensity, scale of operations, market share, industry (where available). Unobserved confounders: Unobserved productivity shocks of the firm.

#	Instrument family (members)	V	N	Data source (link)	How to get it
1	Project Timeline Requirements (+2: Project Timeline Targets; Project Timelines)	3	2	contractsfinder.service.gov.uk/apidocumentation	Free OCDS JSON/CSV via Contracts Finder API or daily CSV on data.gov.uk; merge to Companies House Accounts bulk XBRL files (free) from https://download.companieshouse.gov.uk/en_accountsdata.html .
2	Inventory Management	3	2	US Census of Manufactures / ASM	Restricted-use microdata through the Federal Statistical Research Data Centers (FSRDC); apply for access. MOPS is a supplement to the ASM, enabling linkage to ASM/CMF plant-level inputs and outputs.
3	Maintenance Costs	3	2	India Annual Survey of Industries	Apply for access via MOSPI's National Data Archive (registration + request/fee); approved users download unit-level files.
4	Operational Risk Assessment	3	2	data.cms.gov/provider-data/dataset/yq43-i98g	Free public download (CSV via Provider Data Catalog). Merge with HCRIS cost report data (free public files on CMS).
5	Sustainability Goals	3	2	Compustat	SBTi: free .xlsx download from the Target Dashboard; Compustat: WRDS subscription/restricted academic access (Compustat Annual Updates – Fundamentals Annual).
6	Quality Control Costs	3	2	NA	NA
7	Operational Flexibility	3	2	NA	NA

Notes: 9 candidate IV names from the two-step role-play strategy (pooled across runs; raw LLM-judge non-novelty rating at most 2 and validity rating at least 3 on at least one card), grouped into 7 instrument families by the judge pipeline's embedding clustering; further family members are listed in parentheses. V (N) is the maximum validity (minimum non-novelty) rating over the family's novel-and-valid cards. One row per family and distinct data source; a family whose members found several data sources has several rows. NA = no dataset found for any member.

Table A12: Datasets for candidate IV families: Peer Effects on Technology Adoption (Domain 5).

Datasets include these variables: treatment (D) = A farmer’s influence on peers (the peer/reference farmer’s behaviour or adoption); outcome (Y) = Peers’ technology adoption decisions (adoption of a new agricultural technology); covariates (X) = age, gender, crop type (where available). Unobserved confounders: Shared background and preferences that put the farmer and the peers in the same neighbourhood / social network.

#	Instrument family (members)	V	N	Data source (link)	How to get it
1	Educational Workshops Conducted (+1: Workshops Hosted)	3	2	doi:10.71682/10549340	Free download from CIMMYT Dataverse after accepting terms; includes two public data files (household data and lead-farmer data) and questionnaires.
2	Number of Demonstrations Conducted (+1: Number of Technology Demonstrations Conducted)	3	2	Beaman et al. Malawi network targeting	Free download from openICPSR (public-use Stata files: mnw_public.dta, census.dta, conversation.dta, panel_seed.dta, etc.).
	<i>same family, other data source</i>			microdata.worldbank.org/catalog/2285	Free public-use microdata via World Bank Microdata Library (registration/login required to download).
3	Number of Training Sessions Conducted (+1: Training Sessions Conducted)	3	2	Beaman et al. Malawi network targeting	Free download (openICPSR replication package).
	<i>same family, other data source</i>			doi:10.71682/10549340	Free download after accepting custom terms on CIMMYT Dataverse (files are public).
4	Collaborations with Experts	3	2	Beaman et al. Malawi network targeting	Free public download from openICPSR (AEA Data and Code Repository).
5	Number of Demonstration Plots	3	2	microdata.worldbank.org/catalog/2285	Free public-use microdata via the MCC external repository linked on the World Bank Microdata page (no fee/registration typically required).
6	Positive Testimonials Count (+1: Testimonial Count)	3	2	NA	NA
7	Media Presence	3	2	NA	NA
8	Public Speaking Engagements	3	2	NA	NA
9	Successful Case Studies Shared	3	2	NA	NA

Notes: 13 candidate IV names from the two-step role-play strategy (pooled across runs; raw LLM-judge non-novelty rating at most 2 and validity rating at least 3 on at least one card), grouped into 9 instrument families by the judge pipeline’s embedding clustering; further family members are listed in parentheses. V (N) is the maximum validity (minimum non-novelty) rating over the family’s novel-and-valid cards. One row per family and distinct data source; a family whose members found several data sources has several rows. NA = no dataset found for any member.

Table A13: Datasets for candidate IV families: Peer Effects on Teenage Smoking (Domain 6).

Datasets include these variables: treatment (D) = A teenager's influence on peers through social media (the peer's behaviour / influence); outcome (Y) = Peers' smoking decisions; covariates (X) = age, gender, urbanicity, education level, family income level (where available). Unobserved confounders: Shared background and preferences that put the teenager and the peers in the same social network.

#	Instrument family (members)	V	N	Data source (link)	How to get it
1	Frequency of Posts (+3: Post Timing; Content Frequency; Post Frequency)	3	2	Notre Dame NetHealth	Free download (CSV/DTA) from project website; optional form/email registration requested. Codebooks linked on the Documentation page.
2	Platform Diversity (+3: Platform Choice; Platform Used; Social Media Platform Diversity)	3	2	Notre Dame NetHealth	Free download (CSV/DTA) from project site; optional registration form; codebooks provided on the Documentation page.
<i>same family, other data source</i>				keck.usc.edu/cana/social-networks-study	Restricted; contact investigators/USC CANA for access. Related publications note data are available on reasonable request.
3	Type of Content (+3: Content Style; Content Type; Content Variety)	3	2	PATH Study	Restricted-use microdata via ICPSR/NAHDAP; application and data use agreement required (no fee).
4	Engagement Rate (+2: Peer Interaction Rate; Video Content Engagement)	3	2	Notre Dame NetHealth	Free download (CSV/DTA + PDF codebooks) after completing a short notification form; data files are hosted via public Google Drive links.
5	Content Virality (+2: Content Virality Score; Content Reach)	3	2	Add Health	Restricted-use contract via UNC CPC Data Portal (network identifiers required); public-use files downloadable via ICPSR. Apply for restricted-use to link friends and construct network measures.

continued on next page

#	Instrument family (members)	V	N	Data source (link)	How to get it
6	Content Quality (+1: Content Quality Score)	3	2	Notre Dame NetHealth	Free download (CSV/DTA) from project site via Google Drive; optional short form to notify usage.
7	Influence Score	3	2	Notre Dame NetSense	Free public download of de-identified CSV/R/Stata files from the project’s download page.
8	Hashtag Usage (+4: Hashtag Strategy; Hashtag Use; Use of Hashtags; Use of Influential Hashtags)	3	2	NA	NA
9	Sentiment Analysis (+2: Sentiment Analysis Score; Sentiment of Comments)	3	2	NA	NA
10	Ad Spend (+1: Social Media Advertising)	3	2	NA	NA
11	Influencer Collaborations (+1: Influencer Status)	3	2	NA	NA
12	Post Aesthetics Quality (+1: Visual Appeal)	3	2	NA	NA
13	Number of Followers	3	2	NA	NA
14	Story Views	3	2	NA	NA

Notes: 37 candidate IV names from the two-step role-play strategy (pooled across runs; raw LLM-judge non-novelty rating at most 2 and validity rating at least 3 on at least one card), grouped into 14 instrument families by the judge pipeline’s embedding clustering; further family members are listed in parentheses. V (N) is the maximum validity (minimum non-novelty) rating over the family’s novel-and-valid cards. One row per family and distinct data source; a family whose members found several data sources has several rows. NA = no dataset found for any member.

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